A PROJECT REPORT ON

THE TRANSFORMATIVE ROLE OF AI IN MEDICAL EDUCATION: ENHANCING LEARNING AND TRAINING

Submitted By

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in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE & ENGINEERING

Under the Esteemed guidance of

K. GOWTHAM RAJU M.Tech

(Assistant Professor)

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

ELURU COLLEGE OF ENGINEERING AND TECHNOLOGY

DUGGIRALA (V), PEDAVEGI (M), ELURU-534004

(AFFLICATED TO JNTUK, KAKINADA, ANDHRA PRADESH (INDIA))

2021-2025

DECLARATION

We here by declaring that the project entitled **"THE TRANSFORMATIVE ROLE OF AI IN MEDICAL EDUCATION: ENHANCING LEARNING AND TRAINING"** submitted for the B.Tech. (CSE) degree is our original work and the project has not formed the basis for the award of any other degree, diploma, fellowship or any other similar titles.

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Place :

Date :

CERTIFICATE

This is to certify that the Project titled "**THE TRANSFORMATIVE ROLE OF AI IN MEDICAL EDUCATION: ENHANCING LEARNING AND TRAINING**" is the bonafide work carried out by **K. ANIL KUMAR (21JD1A0548), M. ABHISHEK KUMAR (21JD1A0557), CH. L. S. S. VIGNESH 21JD1A0521, M NAGA MANINDRA (21JD1A0555)** students of B.Tech (CSE) of Eluru College of Engineering and Technology, affiliated to JNT University, Kakinada, AP (India) during the academic year 2024-25, in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (Computer Science and Engineering) and that the project has not formed the basis for the award previously of any other degree, diploma, fellowship or any other similar title.

Signature of the Guide	**Signature of the HOD**
K. GOWTHAM RAJU M.Tech	**Dr. S. SURESH , B. Tech , M. Tech, Ph.D**
Assistant Professor	**Professor**

EXTERNAL EXAMINER

VISION – MISSION - PEOs

Institute Vision	To Achieve Excellence in Engineering Education
Institute Mission	IM1: To deliver quality education through good infrastructure, facilities and committed staff. IM2: To train students as proficient, competent and socially responsible engineers. IM3: To promote research and development activities among faculty and students for betterment of the society.
Department Vision	Empower the students of computer science and engineering department to be technologically strong, innovative and global citizens maintaining human values.
Department Mission	DM 1: Inspire students to become self-motivated and problem solving individuals. DM 2: Furnish students for professional career with academic excellence and leadership skills. DM 3: Create centre of excellence in Computer Science and Engineering. DM 4: Empower the youth and rural communities with computer education.
Program Educational Objectives (PEOs)	Graduates of Computer Science & Engineering will: PEO1: Excel in Professional career through knowledge in mathematics and engineering principles. PEO2: Able to pursue higher education and research. PEO3: Communicate effectively, recognize, and incorporate societal needs in their professional endeavors. PEO4: Adapt to technological advancements by continuous learning.

PROGRAM OUTCOMES

1	**Engineering Knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
2	**Problem analysis**: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3	**Design/development of solutions**: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations
4	**Conduct investigations of complex problems**: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5	**Modern tool usage**: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6	**The engineer and society**: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7	**Environment and sustainability**: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8	**Ethics**: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9	**Individual and team work**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10	**Communication**: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11	**Project management and finance**: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12	**Life-long learning**: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES

PSO1: Have proficiency in algorithms, networking, s/w Engineering and Web Applications for systematic design of computer based system of varying complexity.

PSO2: Qualifying national, international level competitive examinations for successful higher studies and employment.

PROJECT MAPPINGS

Batch No:	A-15
Project Title	THE TRANSFORMATIVE ROLE OF AI IN MEDICAL EDUCATION: ENHANCING LEARNING AND TRAINING
Project Domain	Web Application
Type of the Project	Application – oriented
Guide Name	K. GOWTHAM RAJU
Student Roll No	**Student Name**
21JD1A0548	K. ANIL KUMAR
21JD1A0557	M. ABHISHEK KUMAR
21JD1A0521	CH. L. S. S. VIGNESH
21JD1A0555	M. NAGA MANINDRA

COURSE OUTCOMES : At the end of the Course/Subject, the students will be able to

CO.No	Course Outcomes (COs)	POs	PSOs	Blooms Taxonomy & Level
R20C501.1	Identify the real world problem with a set of requirements to design a solution.	1,2,3,4,5,6,9,10,11,12	1,2	Level-2: Applying
R20C501.2	Implement, Test and Validate the solution against the requirements for a given problem.	1,2,3,4,5,7,8,9,10,11,12	1,2	Level-3: Analysing
R20C501.3	Lead a team as a responsible member in developing software solutions for real world problems and societal issues with ethics.	1,3,4,5,6,7,8,9,10,12	1,2	Level-4: Analysing
R20C501.4	Participate in discussions to bring technical and behavioral ideas for good solutions.	1,2,3,4,5,6,7,8,9,10,11, 12	1,2	Level-5: Evaluating
R20C501.5	Express ideas with good communication skills during presentations.	1,5,6,7,8,9,10,12	1,2	Level-1: Creating
R20C501.6	Learn new technologies to contribute in the software industry for optimal solutions.	1,2,3,4,5,7,8,9,10,11,12	1,2	Level-6: Creating

Course Outcomes vs PO`s Mapping:

Courses Out Comes	P01	P02	P03	P04	P05	P06	P07	P08	P09	P10	P011	P012
R20C501.1	3	3	3	2	3	3	-	3	3	3	3	3
R20C501.2	3	3	3	3	3	-	3	3	3	3	3	3
R20C501.3	2	-	3	3	3	3	3	3	3	3	3	3
R20C501.4	3	3	3	3	2	3	3	3	3	3	3	3
R20C501.5	3	-	-	-	3	3	3	3	3	3	3	3
R20C501.6	2	3	3	3	3	-	3	3	3	3	3	3
Total	16	12	15	14	17	12	15	18	18	18	18	18
Average	2.6	2	2.6	2.3	2.8	2	3	3	3	3	3	3

Justification of Mapping of Course Outcomes with Program Outcomes:

POs mapped with CO.No:R20C501.1:

PO-1: Engineering Knowledge:

Applying the various concepts learned from engineering to gather requirements**.**

PO-2: Problem Analysis:

Identify the formulae and obtaining the layout of the design using mathematical principles.

PO-3: Design/Development of solutions:

Identifying and Designing solutions for the problems related to health, cultural, society and environmental considerations.

PO-4: Conduct investigation of complex problems:

Researching various journals and published papers and knowing how to solve complex problems.

PO-5: Modern Tool Usage:

Selecting and Implementing the latest versions of the applications for modelling the complexity of the project.

PO-6: The Engineer and Society:

Implementing this knowledge to assess health and safety considerations.

PO-8: Ethics:

Implementing professional ethics and norms in engineering practices.

PO-9: Individual and Teamwork:

Individuals researched on the interested domain and concluded the requirements as a team.

PO-10: Communication:

Communication is needed to identify the problem and work as a team to define a solution.

PO-11: Project Management and Finance:

Implementing the engineering principles to work on complex problems.

PO-12: Life-long Learning:

Recognise the importance of technical knowledge in future studies

POs mapped with CO.No:R20C501.2:

PO-1: Engineering Knowledge:

Implementing the various concepts learned from engineering to gather requirements and design and test a solution.

PO-2: Problem Analysis:

Identify the formulae and obtaining the layout of the design and validate using mathematical principles.

PO-3: Design/Development of solutions:

Identifying and Designing solutions for the problems related to health, cultural, society and environmental considerations.

PO-4: Conduct investigation of complex problems:

Researching various journals and published papers and knowing how to solve complex problems.

PO-5: Modern Tool Usage:

Selecting and Implementing the latest versions of the applications for modelling the complexity of the project.

PO-7: Environment and Sustainability

Implementing this knowledge to assess its affects towards environment.

PO-8: Ethics:

Implementing professional ethics and norms in engineering practices.

PO-9: Individual and Teamwork:

Individuals researched on the interested domain and concluded the requirements as a team.

PO-10: Communication:

Communication is needed to Recognise the problem and work as a team to define a solution.

PO-11: Project Management and Finance:

Implementing the engineering principles to work on complex problems.

PO-12: Life-long Learning:

Recognise the importance of technical knowledge in future studies.

POs mapped with CO.No:R20C501.3:

PO-1: Engineering Knowledge:

Implementing the various concepts learned from engineering to gather requirement and work as a team to design and test a solution.

PO-3: Design/Development of solutions:

Recognising and Designing solutions for the problems related to health, cultural, society and environmental considerations.

PO-4: Conduct investigation of complex problems:

Researching various journals and published papers and knowing how to solve complex problems.

PO-5: Modern Tool Usage:

Selecting and Implementing the latest versions of the applications for modelling the complexity of the project.

PO-6: The Engineer and Society:

Implementing this knowledge to assess health and safety considerations.

PO-7: Environment and Sustainability

Implementing this knowledge to assess its affects towards environment.

PO-8: Ethics:

Implementing professional ethics and norms in engineering practices.

PO-9: Individual and Teamwork:

Individuals researched on the interested domain and concluded the project as a team.

PO-10: Communication:

Communication is needed to Recognise the problem and work as a team to obtain a solution.

PO-12: Life-long Learning:

Recognize the importance of technical knowledge in future studies.

POs mapped with CO.No:R20C501.4:

PO-1: Engineering Knowledge:

Implementing the various concepts learned from engineering to gather requirement and work as a team to design and test a solution.

PO-2: Problem Analysis:

Recognise the formulae and obtaining the layout of the design and validate using mathematical principles.

PO-3: Design/Development of solutions:

Recognising and Designing solutions for the problems related to health, cultural, society and environmental considerations.

PO-4: Conduct investigation of complex problems:

Researching various journals and published papers and knowing how to solve complex problems.

PO-5: Modern Tool Usage:

Selecting and Implementing the latest versions of the applications for modelling the complexity of the project.

PO-6: The Engineer and Society:

Implementing this knowledge to assess health and safety considerations.

PO-7: Environment and Sustainability

Implementing this knowledge to assess its affects towards environment.

PO-8: Ethics:

Implementing professional ethics and norms in engineering practices.

PO-9: Individual and Teamwork:

Individuals researched on the interested domain and concluded the project as a team.

PO-10: Communication:

Communication is needed to Recognise the problem and work as a team to obtain a solution.

PO-11: Project Management and Finance:

Implementing the engineering principles to work on complex problems.

PO-12: Life-long Learning:

Recognize the importance of technical knowledge in future studies.

POs mapped with CO.No:R20C501.5:

PO-1: Engineering Knowledge:

Implementing the various concepts learned from engineering to gather requirement and work as a team to design and test a solution.

PO-5: Modern Tool Usage:

Selecting and Implementing the latest versions of the applications for modelling the complexity of the project.

PO-6: The Engineer and Society:

Implementing this knowledge to assess health and safety considerations.

PO-7: Environment and Sustainability

Implementing this knowledge to assess its affects towards environment.

PO-8: Ethics:

Implementing professional ethics and norms in engineering practices.

PO-9: Individual and Teamwork:

Individuals researched on the interested domain and concluded the project as a team.

PO-10: Communication:

Communication is needed to Recognise the problem and work as a team to obtain a solution.

PO-12: Life-long Learning:

Recognize the importance of technical knowledge in future studies.

POs mapped with CO.No:R20C501.6:

PO-1: Engineering Knowledge:

Implementing the various concepts learned from engineering to gather requirement and work as a team to design and test a solution.

PO-2: Problem Analysis:

Recognise the formulae and obtaining the layout of the design and validate using mathematical principles.

PO-3: Design/Development of solutions:

Identifying and Designing solutions for the problems related to health, cultural, society and environmental considerations.

PO-4: Conduct investigation of complex problems:

Researching various journals and published papers and knowing how to solve complex problems.

PO-5: Modern Tool Usage:

Selecting and Implementing the latest versions of the applications for modelling the complexity of the project.

PO-7: Environment and Sustainability

Implementing this knowledge to assess its affects towards environment.

PO-8: Ethics:

Implementing professional ethics and norms in engineering practices.

PO-9: Individual and Teamwork:

Individuals researched on the interested domain and concluded the project as a team.

PO-10: Communication:

Communication is needed to Recognise the problem and work as a team to obtain a solution.

PO-11: Project Management and Finance:

Implementing the engineering principles to work on complex problems.

PO-12: Life-long Learning:

Recognize the importance of technical knowledge in future studies.

Course Outcomes vs PSOs Mapping:

Courses Out Comes	PSO1	PSO2
R20C501.1	3	3
R20C501.2	3	3
R20C501.3	3	3
R20C501.4	3	3
R20C501.5	3	3
R20C501.6	3	3
Total	18	18
Average	3	3

Justification of Mapping of Course Outcomes with Program Specific Outcomes:

PSOs mapped with CO.No:R20C501.1:

PS0-1: Gaining knowledge in software engineering practices to gather requirements.

PSO-2: Able to use this knowledge in further studies.

PSOs mapped with CO.No:R20C501.2:

PS0-1: Gaining knowledge in software engineering practices like algorithms to design a solution.

PSO-2: Able to use this knowledge in further studies.

PSOs mapped with CO.No:R20C501.3:

PS0-1: Gaining knowledge in software engineering practices to gather requirements and networking for designing and communication.

PSO-2: Able to use this knowledge in further studies.

PSOs mapped with CO.No:R20C501.4:

PS0-1: Gaining knowledge in software engineering practices to gather requirements and bring in technical and behavioural ideas.

PSO-2: Able to use this knowledge in further studies.

PSOs mapped with CO.No:R20C501.5:

PS0-1: Gaining knowledge in software engineering practices to gather requirements and using good communication skills to express ideas.

PSO-2: Able to use this knowledge in further studies.

PSOs mapped with CO.No:R20C501.6:

PS0-1: Gaining knowledge in software engineering practices to learn to explore new technologies.

PSO-2: Able to use this knowledge in further studies.

Mapping Level	**Mapping Description**
1	Low Level Mapping with PO & PSO
2	Moderate Mapping with PO & PSO
3	High Level Mapping with PO & PSO

Member 1: K. ANIL KUMAR (21JD1A0548)

Member 2:M.ABHISHEK KUMAR(21JD1A0557)

Member 3: CH. L. S. S. VIGNESH (21JD1A0521)

Member 4: M. NAGA MANINDRA (21JD1A0555)

PROJECT GUIDE

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First, we would like to express our deepest appreciation to our advisor and guide, **Mr. K. GOWTHAM RAJU** for his sincere guidance, heart full encouragement, and continuous support during our B. Tech project at Eluru College of Engineering and Technology. His vision for insights into the problem resolution has greatly inspired us to put things in the right direction. It has been a great pleasure to conduct our project under his guidance.

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ABSTRACT

With rapid advancements in science and technology, Artificial Intelligence (AI) has emerged as a transformative force in medical education. AI is revolutionizing various aspects of learning and training by enhancing medical simulations, personalized tutoring and decision-support systems. This article explores the applications of AI in medical education, including virtual patient simulations, intelligent tutoring systems, adaptive learning platforms. Our findings indicate that AI enhances medical instruction by improving efficiency and replicating expert decision-making to create personalized learning experiences. Its impact is particularly significant in fostering clinical reasoning, skill development, and medical innovation. The growing integration of AI in medical curricula underscores its crucial role in shaping the future of healthcare education and training.

LIST OF FIGURES

CONTENTS

1. INTRODUCTION

As humans place great hopes on artificial intelligence, artificial intelligence can have bright prospects, and it may be applied to all aspects of our lives to improve the overall standard of living of all of us. With the development of artificial intelligence technology, scientists have gradually applied artificial intelligence technology to the teaching field, and quite good results have been achieved. Therefore, the development and progress in artificial intelligence, combined with teaching, will be an excellent new teaching method.

Artificial intelligence is a new science that emerged in the middle of the 20th century. This science mainly belongs to computer science, but it covers information science, linguistics, psychology, philosophy, mathematics, and many other disciplines. It is a discipline that has strong comprehensiveness. Artificial intelligence mainly uses computer systems to simulate human thinking activities. This discipline has a wide research scope and it has also been applied in many aspects. Because artificial intelligence has wide research fields, it is also a very challenging science category requiring scientists to have a strong knowledge base in all aspects. At present, the research of artificial intelligence is closely related to the current needs of human beings. The research on artificial intelligence technology has also evolved with the changes of the times, so that the artificial intelligence technology can be applied to more meaningful things. The main goal of artificial intelligence is to require computers to have “abilities to acquire and learn knowledge”, “abilities to process knowledge”, “abilities to understand language”, “the ability to infer and search automatically”, and abilities in many other aspects. In terms of research objects, artificial intelligence can be divided into three different areas. The first one is the ability of “natural language processing” and to write computer programs that can be read and spoken. The second one is to develop a machine that has sensitive sensory, and can simulate human hearing and vision and distinguish different environments automatically. The third type is an R&D expert system that uses a computer to simulate an expert’s behavior.

In terms of the research nature of artificial intelligence, it can be divided into two aspects: theory and engineering. Theoretical research is the continuous development and expansion of artificial intelligence theory. Engineering research is to design and develop corresponding products. These two aspects are closely connected and indivisible. Theoretical research provides a theoretical basis for engineering research; engineering research applies theoretical research to practice. Artificial intelligence has the following technical characteristics: search ability, knowledge expression function, reasoning ability, abstraction ability, speech recognition ability,

ability to process fuzzy information. These five points have basically made it possible for artificial intelligence to simply simulate human thinking In the past decade, the application of artificial intelligence has solved or partially solved many challenges in the education field, including language processing, reasoning, planning and cognitive modeling. Artificial intelligence provides students with more opportunities to participate in a digital and dynamic way. These opportunities are often not found in outdated textbooks or the fixed environment of the classroom. Through this collaborative learning method, each student has the potential to advance others, and can accelerate the exploration of new learning and the creation of innovative technologies. Four applications are provided below to illustrate how artificial intelligence can be applied to medical education. DxR Clinician is an online virtual patient system that uses artificial intelligence technology specifically for teaching hospitals, medical colleges, and residents. The system is widely used in education and clinical thinking evaluation of medical students. The software collects hundreds of real patient data and is compiled by experts and artificial intelligence as specific cases. These cases cover a wide range of clinical issues. Medical students make diagnoses through inquiry, simulated physical examinations, and supplementary examinations of virtual patients to diagnose and provide treatment plans. For teachers, DxR Clinician can be used as a useful analysis tool to help teachers understand students' behavior and adjust courses through appraisal results. For students, they can quickly develop clinical problem solving skills. By interacting with the cases, students can learn a lot about important disease diagnosis. At the same time, the system can identify mistakes that students make in the process of case analysis, conducts deep learning and analysis, and help students solve these problems. One kind of computer software that has similar function with DxR Clinician is called Intelligent Tutor Systems, which can track the learner's "psychological steps" in the process of solving problems to diagnose the wrong concepts and estimate the learners' understanding extent of the field. The Intelligent Tutor System can also provide learners with timely guidance, feedback and explanation, and can promote learners' learning behaviors such as self-regulation, self-monitoring and self-explaining. Distance education is a kind of teaching method that is not limited by time and space and can realize real-time on-line and off-line teaching. Learning, communication and sharing can be conducted through web-based teaching methods such as microblogging; virtual simulation training, mobile ward round in clinical practice teaching and mobile nursing play n important role in medical teaching, especially virtual simulation teaching technology has gained more in-depth and extensive application; the development of remote transmission technology of imaging and pathological films, instant transfer technology, all online storage technology, active monitoring and self-healing technology, integrated platform technology, three-dimensional post-processing, computer-aided diagnosis, and medical imaging real-time conferencing technology have had a profound impact on the teaching methods; regional Picture Archiving and Communication System (PACS) and regional pathology platform.

In the aspects of continuing medical education, China has adopted a dual approval system for institutions and projects. At present, 50 state-level continuing medical education project bases have been approved, and more than 4,000 state-level continuing medical education projects have been newly announced every year. Since the exploration of distance continuing medical education in 1996, the Ministry of Health has successively approved shuangwei net, haoyisheng net, China Stomatology net, Shanghai Zhongshan Hospital, West China Medical Center and Medical Network College of Peking University to carry out distance medical education and assess these institutions in 2006 and 2011. Each year, more than 1500 experts participate in distance continuing medical education covering 20 secondary disciplines and 74 tertiary disciplines. The number of certifications issued by state-level continuing medical education projects in the aforementioned distance learning institutions from 2000 to 2010 is approximately 3 million. This is four times that of the traditional method of education in the same period. Through modern information technology, data centers, teaching resources library, cloud platform, are constructed for students recruiting, training process management and evaluation, which can improve the efficiency and service level of continuing medical education management. In the sharing management of the base, institutional management, trainees, project management, evaluation, credit management and teaching content, modern information technology can be applied, such as the establishment of a continuing education object database, covering the basic information of each student, learning processes and evaluation conditions, and the establishment of a national continuing medical education base and institutional management information system. In 2005, online reporting, online assessment and online publication of national continuing medical education projects were realized. In residency training, in the process of students' recruitment, announcement, acceptance, teachers' teaching and courses set-up, the common parts can be exchanged and coordinated through the computer system. Among base hospitals, health administrative departments, clinical teachers and departments, the same information can be transmitted through information means (web pages, mobile phones). Synchronizing courses through computer information systems can achieve data exchange, information sharing and business collaboration among different courses.

1.1 MOTIVATION

The rapid advancement of Artificial Intelligence (AI) is revolutionizing various fields, and medical education is no exception. Traditional teaching methods in medical training often struggle to keep pace with the growing complexity of healthcare and the increasing need for highly skilled professionals. AI-driven technologies offer innovative solutions to enhance learning experiences, improve clinical decision-making, and bridge gaps in medical education.

With the increasing demand for competent healthcare professionals, there is a pressing need to integrate intelligent learning systems that personalize medical training, automate assessments, and provide data-driven insights into student performance. AI-powered tools such as virtual patient simulations, intelligent tutoring systems, and adaptive learning platforms significantly enhance medical education by offering interactive and realistic learning environments. These technologies enable students to develop critical thinking, diagnostic accuracy, and clinical reasoning skills through hands-on virtual practice.

Furthermore, AI assists educators by automating administrative and repetitive tasks, allowing them to focus on interactive and student-centered teaching approaches. By leveraging AI-driven simulations, real-time feedback mechanisms, and predictive analytics, this research aims to demonstrate how AI can transform medical education, making learning more accessible, efficient, and impactful. As AI continues to evolve, its integration into medical training will play a crucial role in preparing future healthcare professionals with the knowledge and skills required for modern medical practice.

1.2 PROBLEM DEFINITION

Traditional medical education relies heavily on classroom-based instruction, textbooks, and standardized training methods, which often fail to provide personalized and adaptive learning experiences. These conventional approaches limit students' ability to develop critical diagnostic and clinical reasoning skills in dynamic and real-world medical scenarios. Additionally, the lack of interactive and AI-driven tools makes it challenging for students to practice complex medical decision-making in a risk-free environment.

Educators also face difficulties in effectively addressing individual learning needs, providing real-time feedback, and managing large volumes of coursework. Limited student-teacher interaction, coupled with the absence of intelligent automation, increases the workload for instructors and reduces their ability to focus on skill development and mentorship. Moreover, traditional assessment methods do not offer comprehensive insights into students’ cognitive processes, making it difficult to track their progress and areas of improvement.

To address these challenges, this research explores the integration of AI-driven technologies in medical education to enhance learning and training experiences. By leveraging AI-powered virtual patient simulations, intelligent tutoring systems, and real-time feedback mechanisms, the proposed approach aims to create a more interactive, personalized, and efficient medical training system. AI will assist students in developing clinical problem-solving skills, improving diagnostic accuracy, and engaging in adaptive learning tailored to their individual needs. This transformation in medical education will bridge the gap between theoretical knowledge and practical application, ultimately preparing future healthcare professionals more effectively.

1.3 OBJECTIVE OF THE PROJECT

The objective of this project is to develop an AI-powered system that enhances learning and training in medical education. Traditional medical training methods often lack personalized learning features, real-time assistance, and intelligent automation, making it challenging for students to acquire critical diagnostic and clinical decision-making skills effectively.

This project aims to integrate Artificial Intelligence (AI) into medical education to:

- **Enhance Personalized Learning** – Utilize AI-driven adaptive learning systems to tailor medical training content based on individual student performance, clinical reasoning skills, and learning behavior.
- **Provide AI-Powered Assistance** – Develop AI-driven virtual patient simulations and intelligent tutoring systems to offer real-time support, answer medical queries, and guide students through clinical case analyses.
- **Enable Adaptive Content Delivery** – Dynamically adjust and present medical learning materials based on student engagement, knowledge retention, and diagnostic accuracy.

By leveraging Machine Learning (ML), Natural Language Processing (NLP), and AI-driven analytics, this AI-powered system will create a more efficient, interactive, and adaptive learning environment, improving student engagement, clinical competence, and educator productivity in medical education.

2. LITERATURE SURVEY

2.1 Use of Information Technology in Medical Education [J]. Webmed Central Medical Education

Authors: JOSHI S

The past few years have seen rapid advances in communication and information technology (C&IT), and the pervasion of the worldwide web into everyday life has important implications for education. Most medical schools provide extensive computer networks for their students, and these are increasingly becoming a central component of the learning and teaching environment. Such advances bring new opportunities and challenges to medical education, and are having an impact on the way that we teach and on the way that students learn, and on the very design and delivery of the curriculum. The plethora of information available on the web is overwhelming, and both students and staff need to be taught how to manage it effectively. Medical schools must develop clear strategies to address the issues raised by these technologies. We describe how medical schools are rising to this challenge, look at some of the ways in which communication and information technology can be used to enhance the learning and teaching environment, and discuss the potential impact of future developments on medical education.

2 .2 NMC(New Media Consortium) Horizon Report 2016 Higher Education Edition

AUTHORS: Johnson L, Adams Becker S, Cummins M, et al

Nowadays Smart phones have become an irresistible part of everyone's life. With these smart phones the human life is changed for better. "Android Based Campus Solution" is a college management application which basically aimed at managing most of the activities of college. The main objective of this app is to make advancement in education system and institutional activities. This application helps in adding mobility and automation in managing the institutional information. The existing system uses website for publishing notices, or peon helps in circulating notices. This is very time consuming process. The android application simplifies this process by giving instant notifications to the students or concerned staff. This application makes this process easier, faster, secure and less error prone.

3) Learning Environments and Displacement of Learning [J]. British Journal of Educational Technology

AUTHORS : Thomas H. Learning Spaces, Traditionally, at least according to popular wisdom, learning took place in venues that were custom-designed for the purpose. The purpose, given the evidence of the artefacts with which we are confronted, seems to have been the educational equivalent of the production line that so succinctly characterised the industrialisation of society. One consequence of this design logic, however, is that learning is defined as something that is married to a 'place'. This paper will argue that the conceptual 'slippage' that characterises the disappearing differences between 'learning spaces' and 'learning environments', coupled with the further 'displacement' of the learner (turned avatar) in virtual spaces such as *Facebook* and *Second Life,* serves to 'displace' learning itself. The paper argues further that we have failed to recognise the primacy of 'physical situatedness' to our conceptions of learning itself. In short, our difficulty in understanding and articulating the nature of learning is partly brought about by our inability to articulate where learning takes place—in a world characterised by virtual space and electronic selves. If we are to articulate the nature of learning in our age, then we need to articulate the nature of the real and virtual spaces and bodies that we inhabit.

4) Educational Ecosystem in the View of New Industrial Revolution

AUTHORS : Youmei Wang. Cultivating Makers

Higher education in the fourth industrial revolution (HE 4.0) is a complex, dialectical and exciting opportunity which can potentially transform society for the better. The fourth industrial revolution is powered by artificial intelligence and it will transform the workplace from tasks based characteristics to the human centred characteristics. Because of the convergence of man and machine, it will reduce the subject distance between humanities and social science as well as science and technology. This will necessarily require much more interdisciplinary teaching, research and innovation. This paper explores the impact of HE 4.0 on the mission of a university which is teaching, research (including innovation) and service

5) Improving the Quality of University Teaching in Learning Innovation: The Key to Apply Information Technology in Higher Education

AUTHORS: **Xinmin Sang, Yangbin Xie**

Information and communication technologies (ICT) have become commonplace entities in all aspects of life. Across the past twenty years the use of ICT has fundamentally changed the practices and procedures of nearly all forms of endeavour within business and governance. Within education, ICT has begun to have a presence but the impact has not been as extensive as in other

fields. Education is a very socially oriented activity and quality education has traditionally been associated with strong teachers having high degrees of personal contact with learners. The use of ICT in education lends itself to more student-centred learning settings and often this creates some tensions for some teachers and students. But with the world moving rapidly into digital media and information, the role of ICT in education is becoming more and more important and this importance will continue to grow and develop in the 21 st century. This paper highlights the various impacts of ICT on contemporary higher education and explores potential future developments. The paper argues the role of ICT in transforming teaching and learning and seeks to explore how this will impact on the way programs will be offered and delivered in the universities and colleges of the future.

3. SYSTEM ANALYSIS

3.1 EXISTING SYSTEM:

Computer technology could be used to reduce the number of mortality and reduce the waiting time to see the specialist. Computer program or software developed by emulating human intelligence could be used to assist the doctors in making decision without consulting the specialists directly. The software was not meant to replace the specialist or doctor, yet it was developed to assist general practitioner and specialist take an immediate action to produce as many doctors as possible. However, while waiting for students to become doctors and the doctors to become specialists, many patients may already die. Current practice for medical treatment required patients to consult specialist for further treatment. Artificial intelligence provides students with more opportunities to participate in a digital and dynamic way.

Disadvantages of the Existing System

- Lack of Personalization – Limited adaptation to individual patients.
- Data Dependency – Accuracy relies on quality medical data.
- Limited Human Interaction – Lacks empathy and critical thinking.
- High Costs – Expensive to develop and implement.
- Ethical Concerns – Raises data privacy and liability issues.

3.2 PROPOSED SYSTEM

Through modern information technology, data centers, teaching resources library, cloud platform, are constructed for students recruiting, training process management and evaluation, which can improve the efficiency and service level of continuing medical education management. In the sharing management of the base, institutional management, trainees, project management, evaluation, credit management and teaching content such as artificial intelligence in distance medical teaching, virtual inquiry, distance education management, teaching video recording, etc., especially for the improvement of the overall quality of medical students, provide much inspiration for the applications of artificial intelligence in medical education.

Advantages of the Proposed System

1. **Efficient Medical Education Management** – AI-powered data centers and cloud platforms streamline student recruitment, training, and evaluation.

2. **Enhanced Resource Utilization** – Teaching resource libraries and AI-driven content

management improve accessibility to medical learning materials.

3. **Advanced Distance Learning** – AI supports virtual inquiry, remote education, and recorded teaching sessions for flexible learning.
4. **Automated Institutional Management** – AI optimizes trainee tracking, project evaluation, and credit management, reducing administrative workload.
5. **Improved Training Quality** – AI-driven insights and adaptive learning enhance the overall quality of medical education and student performance.

3.3 SYSTEM SPECIFICATION:

HARDWARE REQUIREMENTS:

- ❖ **System** **:** Pentium IV 2.4 GHz or Higher
- ❖ **Hard Disk** **:** 1 TB.
- ❖ **Monitor** : 14’ Colour Monitor.
- ❖ **Mouse** **:** Optical Mouse.
- ❖ **RAM** **:** 4GB.

SOFTWARE REQUIREMENTS:

- ❖ **Operating system** **:** Windows 7 Ultimate
- ❖ **Coding Language** **:** Python (Django Framework)-
- ❖ **Front-End Designing** **:** HTML, CSS, Javascript.
- ❖ **Data Base** **:** MySQL.
- ❖ **Browser** : Any Browser

3.4 FUNCTIONAL REQUIREMENTS

- **User Registration & Authentication** – Secure login for students and educators.
- **AI-Powered Learning Assistance** – AI chatbot helps with medical queries and study materials.
- **Virtual Simulation & Distance Learning** – AI enables virtual patient simulations and remote education.
- **Graphical User Interface (GUI)** – Interactive UI for seamless user experience.

3.5 NON-FUNCTIONAL REQUIREMENTS

- **Usability** – Intuitive and user-friendly design for students and educators.
- **Scalability** – System can handle increasing users and large datasets.
- **Security** – Implements secure authentication and data protection measures.
- **Reliability** – Ensures high uptime and consistent system performance.

Advantages

- Enhances student engagement through AI-driven personalized learning.
- Provides automated grading, reducing instructor workload.
- AI-powered insights improve decision-making in medical education.

Disadvantages

- Requires high computational resources for AI processing.
- AI training takes time and needs a large dataset.
- Implementation complexity is higher than traditional education systems.

3.6 SYSTEM STUDY

3.6.1 FEASIBILITY STUDY

The feasibility of the project is analyzed in this phase and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.

Three key considerations involved in the feasibility analysis are,

- ECONOMICAL FEASIBILITY
- TECHNICAL FEASIBILITY
- SOCIAL FEASIBILITY

3.6.1.1 ECONOMICAL FEASIBILITY

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

3.6.1.2 TECHNICAL FEASIBILITY

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands on the available technical resources. This will lead to high demands being placed on the client. The developed system must have a modest requirement, as only minimal or null changes are required for implementing this system.

3.6.1.3 SOCIAL FEASIBILITY

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system.

3.7 MODULES:

- TRAINER
  - Trainer Home
  - View Queries
  - Upload files
  - Upload Patient Details
- MEDICAL STUDENT
  - Student Home
  - Materials
  - Send Queries
  - View Patient Details
  - Chat with AI
- ADMIN
  - Home
  - Trainers
  - Students
  - View Patient Details
- ARTIFICIAL INTELLIGENCE

3.7.1 TRAINER

In residency training, in the process of students' recruitment, announcement, acceptance, teachers' teaching and courses set-up, the common parts can be exchanged and coordinated through the computer system. Among base hospitals, health administrative departments, clinical teachers and departments, the same information can be transmitted through information means help students at different levels, and provide help and support when teachers and students need it. Artificial intelligence can not only help teachers and students design courses that meet their needs, but also can focus on student performance and alert teachers when problems may arise, helping teachers improve teaching methods. Artificial intelligence will change the role of teachers. Teachers will supplement artificial intelligence courses to provide students with interpersonal interaction and practical experience.

3.7.2 MEDICAL STUDENT

Artificial intelligence provides students with more opportunities to participate in a digital and dynamic way. These opportunities are often not found in outdated textbooks or the fixed environment of the classroom. Through this collaborative learning method, each student has the potential to advance others, and can accelerate the exploration of new learning and the creation of innovative technologies. Four applications are provided below to illustrate how artificial intelligence can be applied to medical education. Virtual Inquiry System, Medical Distance Learning, Influence of Artificial Intelligence Technology on the Management of Distance Medical Education, Recording teachers' teaching videos.

3.7.3 ADMIN:

Admin gives the activation permission to users , admin can verify the trainer and student status. the admin can calculate raw SVM support vector machine. the admin can also calculate the raw svm DT (decision tree algorithm). Window-based anomaly detection divides time-series instances into overlapping windows. Anomaly scores are first calculated per window, and then aggregated for comparison with a predefined threshold. In supervised prediction-based anomaly detection, a machine-learning-based predictive model is first learned from historical logs The accumulated difference between these predicted and the actual observations is defined as the anomaly score for each test time-series instance.

3.7.4 ARTIFICIAL INTELLIGENCE

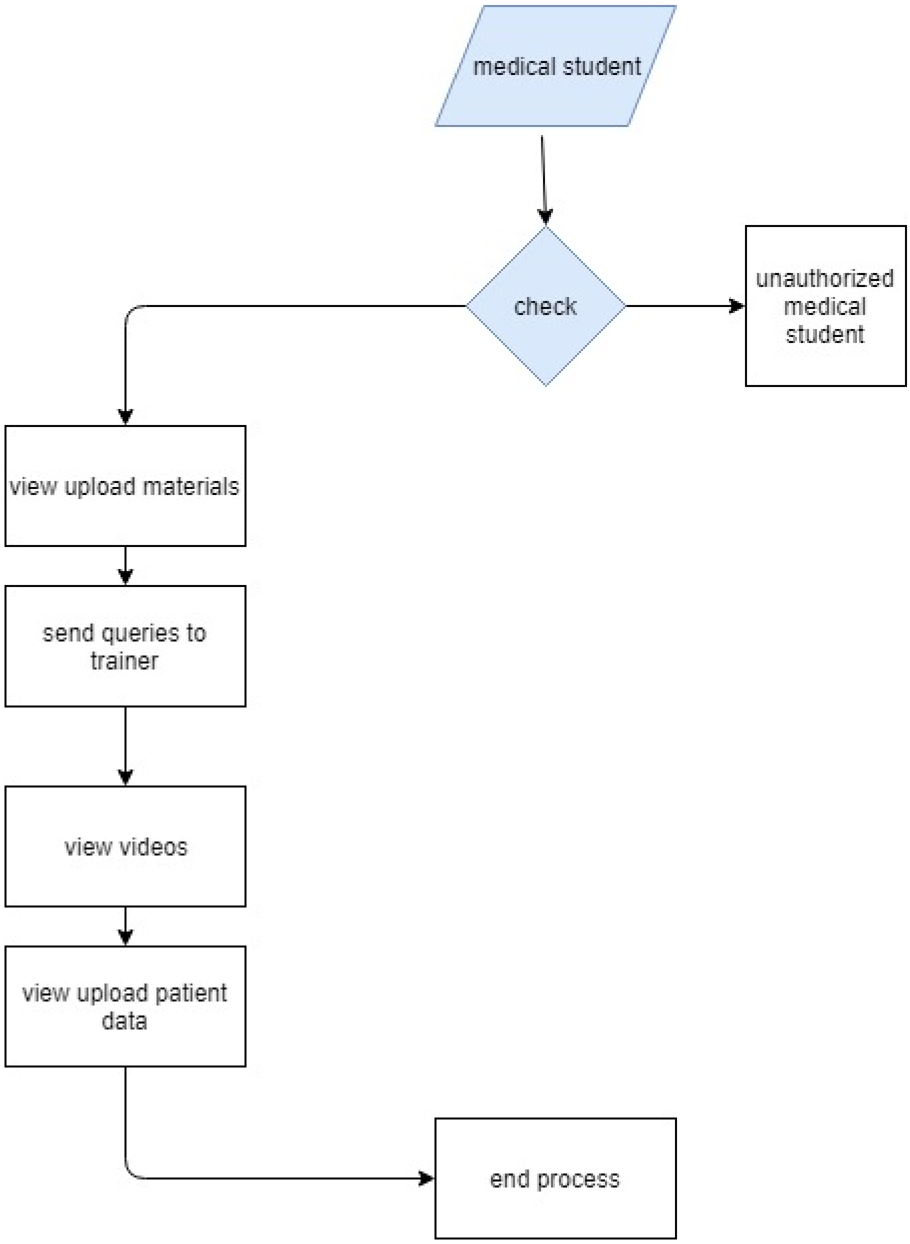
Artificial intelligence is a new science that emerged in the middle of the 20th century. This science mainly belongs to computer science, but it covers information science, linguistics, psychology, philosophy, mathematics, and many other disciplines. It is a discipline that has strong comprehensiveness. Artificial intelligence mainly uses computer systems to simulate human thinking activities. This discipline has a wide research scope and it has also been applied in many aspects. Because artificial intelligence has wide research fields, it is also a very challenging science category requiring scientists to have a strong knowledge base in all aspects. At present, the research of artificial intelligence is closely related to the current needs of human beings. The research on artificial intelligence technology has also evolved with the changes of the times, so that the artificial intelligence technology can be applied to more meaningful things. The main goal of artificial intelligence is to require computers to have "abilities to acquire and learn knowledge", "abilities to process knowledge", "abilities to understand language", "the ability to infer and search automatically", and abilities in many other aspects.

4. SYSTEM DESIGN

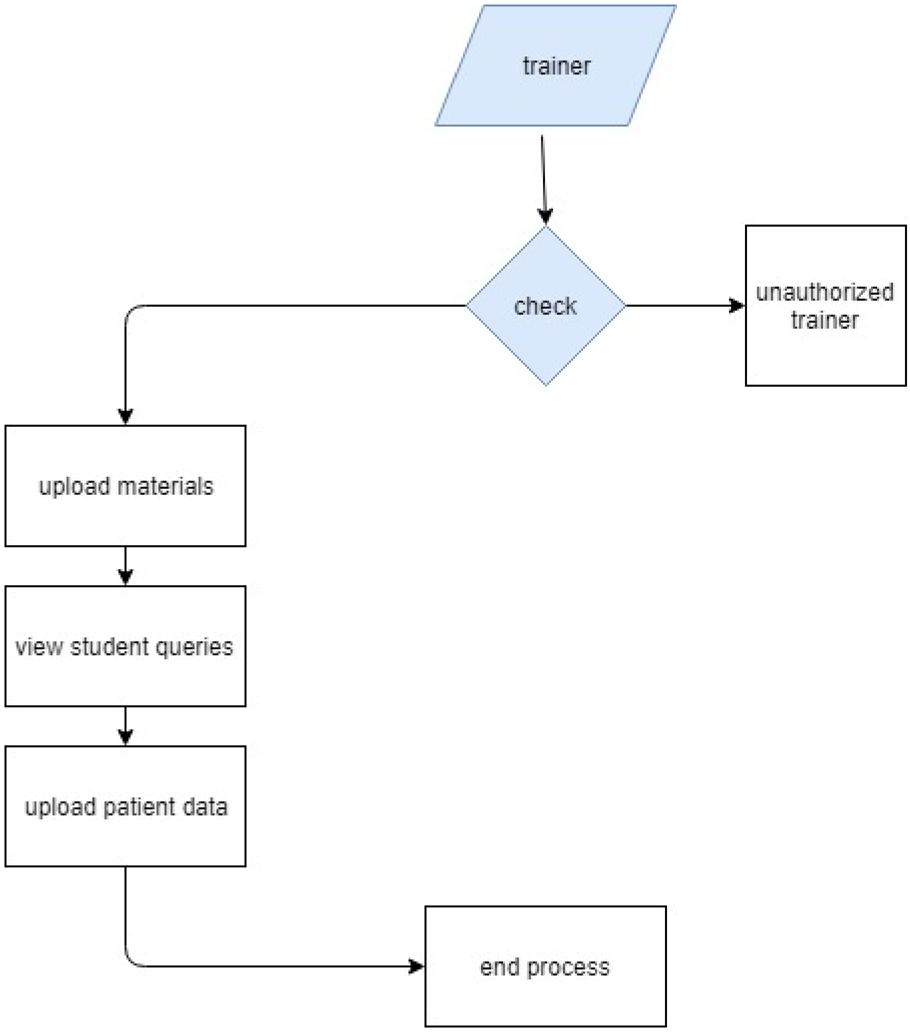
4.1 SYSTEM ARCHITECTURE:

DATA FLOW DIAGRAM:

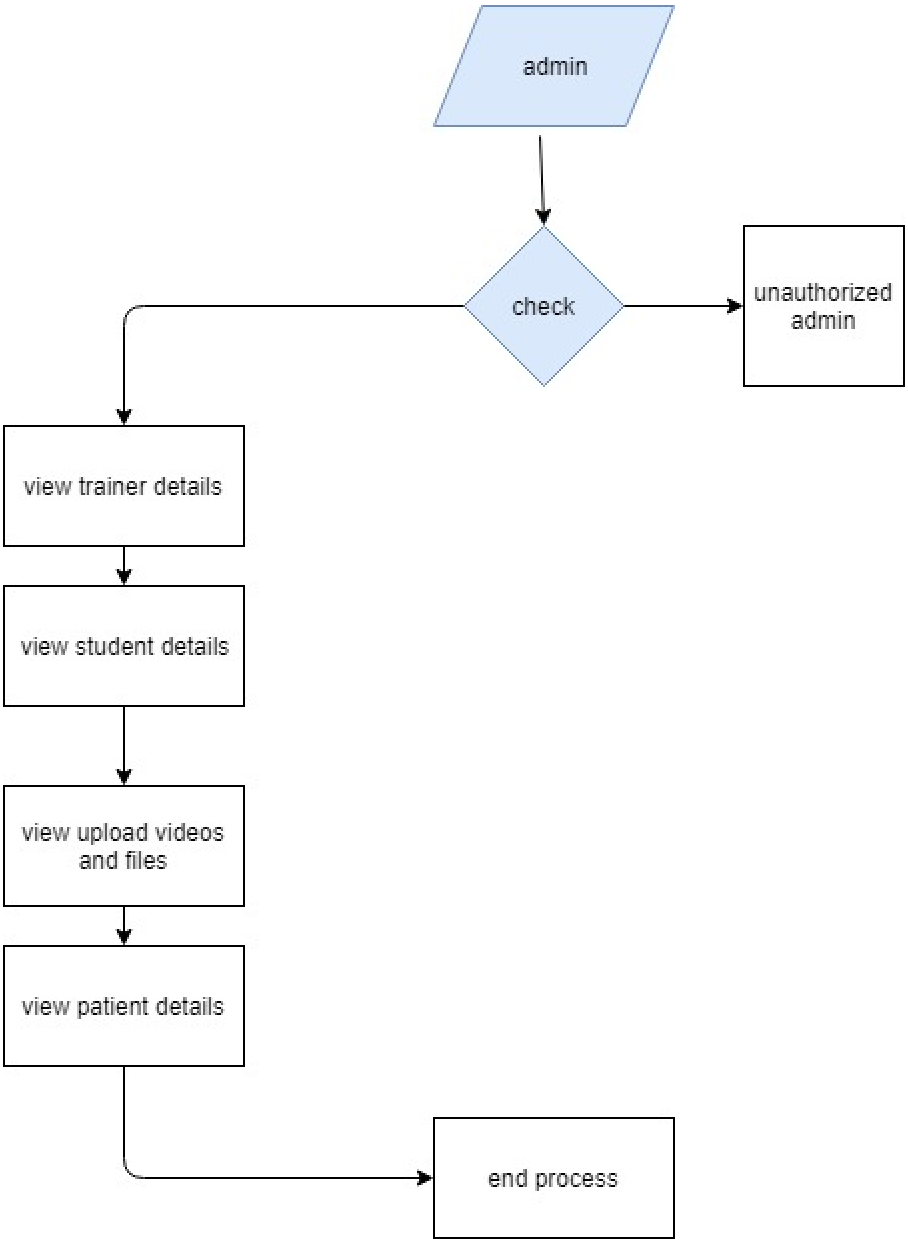
1. The DFD is also called as bubble chart. It is a simple graphical formalism that can be used to represent a system in terms of input data to the system, various processing carried out on this data, and the output data is generated by this system.
2. The data flow diagram (DFD) is one of the most important modeling tools. It is used to model the system components. These components are the system process, the data used by the process, an external entity that interacts with the system and the information flows in the system.
3. DFD shows how the information moves through the system and how it is modified by a series of transformations. It is a graphical technique that depicts information flow and the transformations that are applied as data moves from input to output.
4. DFD is also known as bubble chart. A DFD may be used to represent a system at any level of abstraction. DFD may be partitioned into levels that represent increasing information flow and functional detail.



4.1.1 DFD for Trainer



4.1.2 DFD for Medical Student



4.1.3 DFD for Admin

5.

UML stands for Unified Modeling Language. UML is a standardized general-purpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group.

The goal is for UML to become a common language for creating models of object oriented computer software. In its current form UML is comprised of two major components: a Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML.

The Unified Modeling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modeling and other non-software systems.

The UML represents a collection of best engineering practices that have proven successful in the modeling of large and complex systems.

The UML is a very important part of developing objects oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

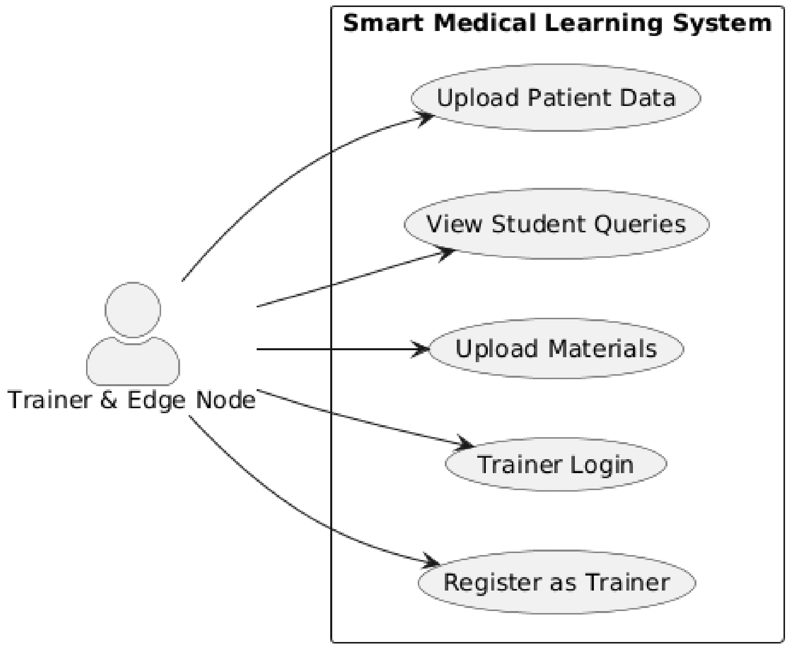
GOALS:

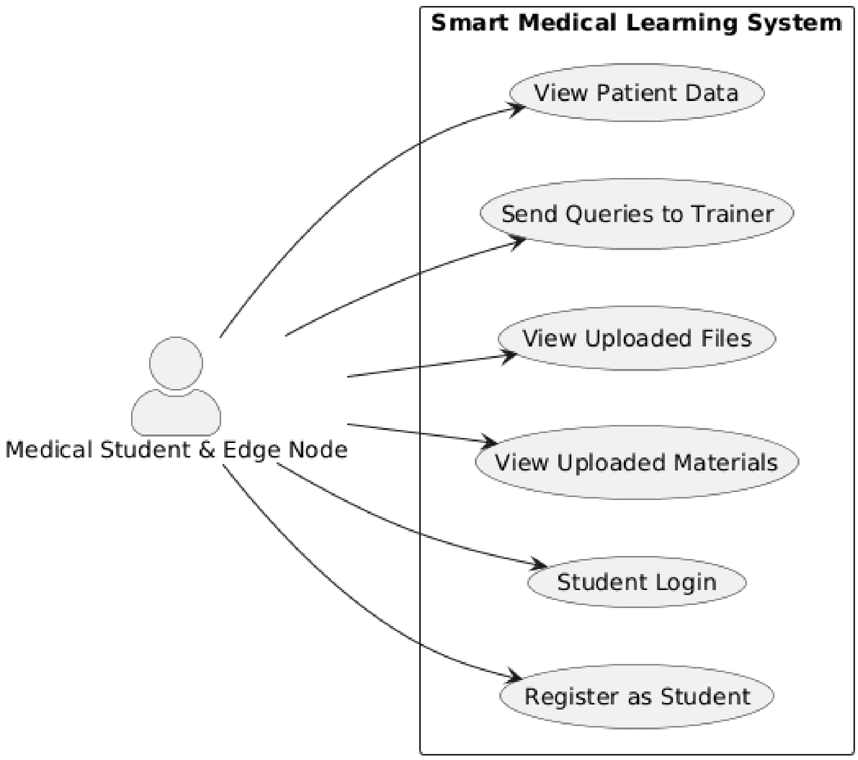
The Primary goals in the design of the UML are as follows:

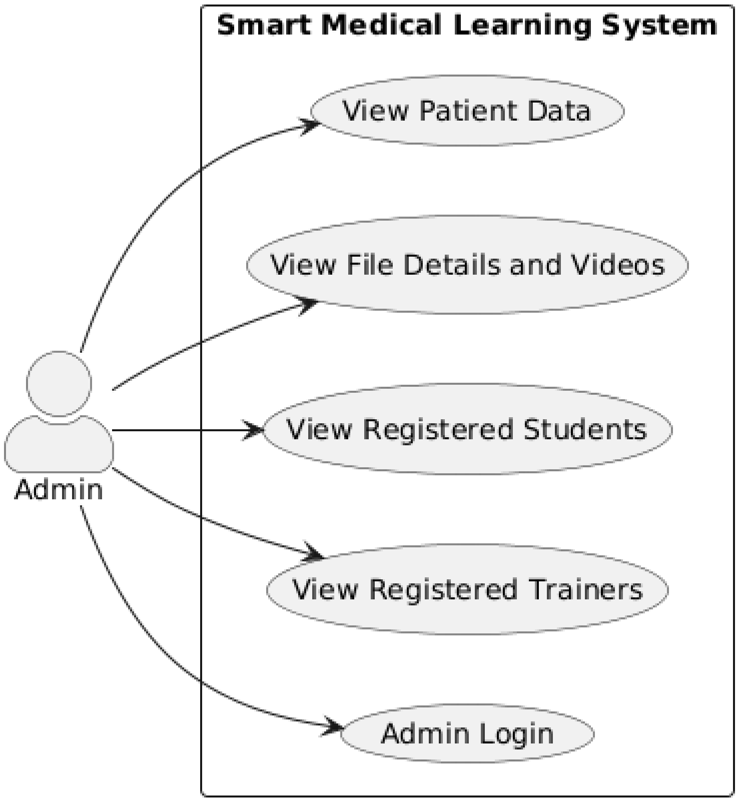
1. Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.
2. Provide extendibility and specialization mechanisms to extend the core concepts.
3. Be independent of particular programming languages and development process.
4. Provide a formal basis for understanding the modeling language.
5. Encourage the growth of OO tools market.
6. Support higher level development concepts such as collaborations, frameworks, patterns and components.
7. Integrate best practices.

4.2.1 USE CASE DIAGRAM:

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.


Smart Medical Learning System
Upload Patient Data
View Student Queries
Upload Materials
Trainer Login
Register as Trainer
Trainer & Edge Node

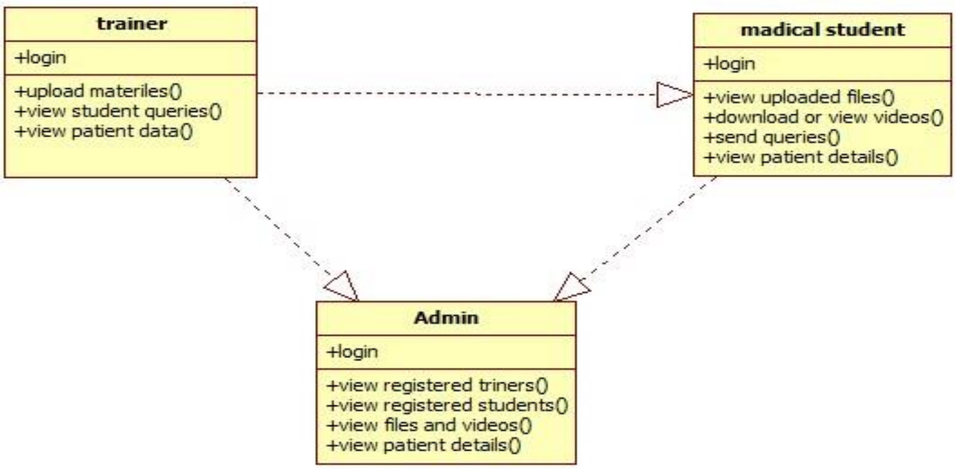

Smart Medical Learning System
View Patient Data
Send Queries to Trainer
View Uploaded Files
View Uploaded Materials
Student Login
Register as Student
Medical Student & Edge Node



4.2.1 **Use case Diagram**

4.2.2 CLASS DIAGRAM:

In software engineering, a class diagram in the Unified Modeling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among the classes. It explains which class contains information.

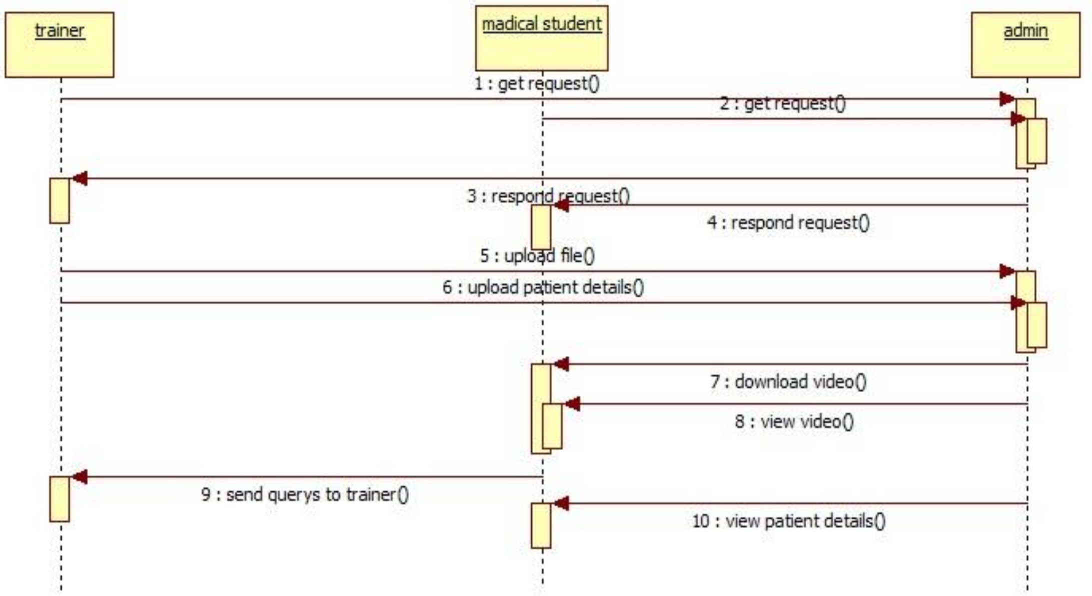


4.2.2 Class Diagram

4.2.3 SEQUENCE DIAGRAM:

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams.

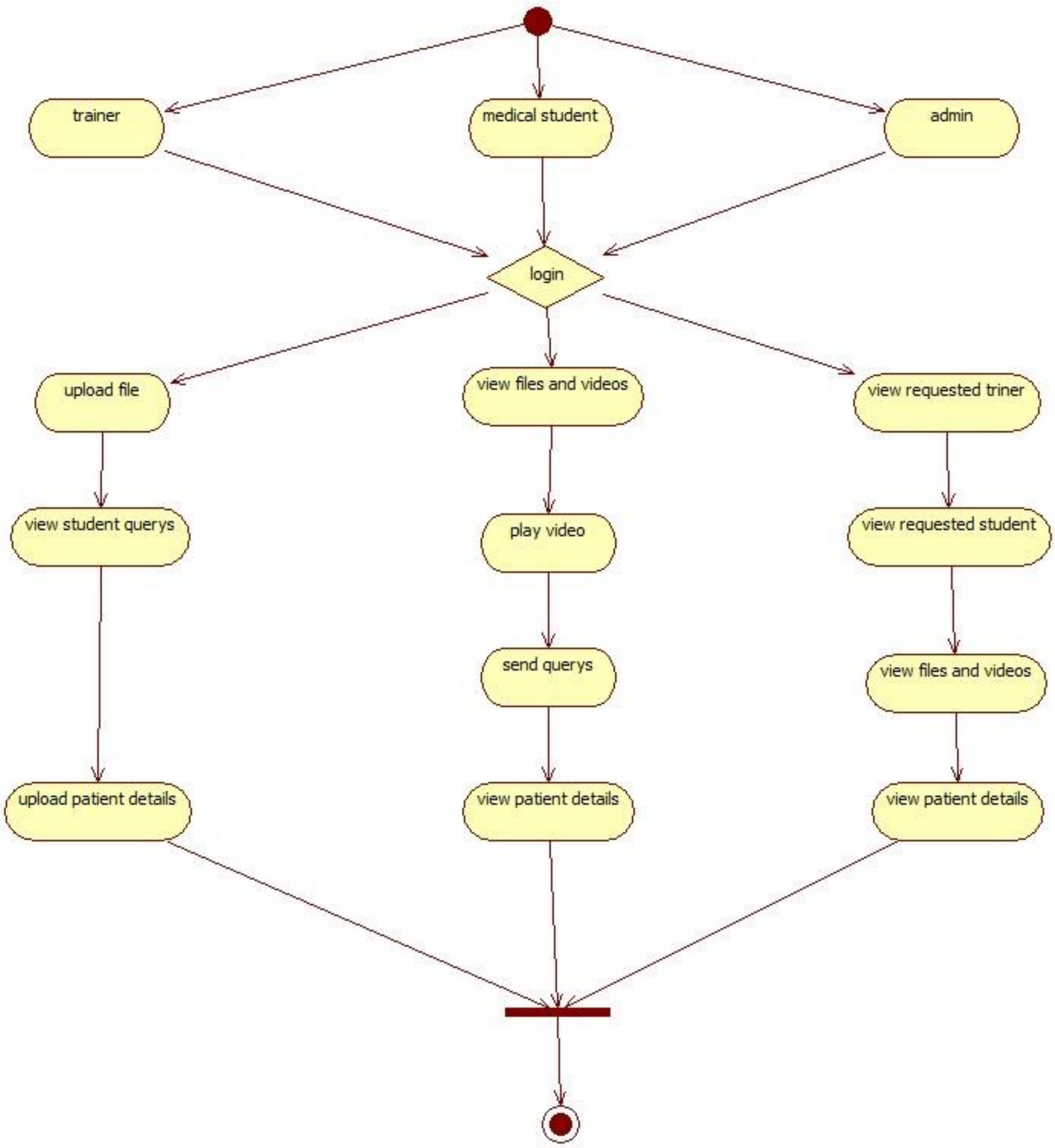
4.2.3



Sequence Diagram

4.2.4 ACTIVITY DIAGRAM:

Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. In the Unified Modeling Language, activity diagrams can be used to describe the business and operational step-by-step workflows of components in a system. An activity diagram shows the overall flow of control.



4.2.4 Activity Diagram

5. SYSTEM IMPLEMENTATION

5.1 SOURCE CODE

Urls.py

```
"""MedicalEducation URL Configuration


The `urlpatterns` list routes URLs to views. For more information please see:
    https://docs.djangoproject.com/en/2.2/topics/http/urls/
Examples:
Function views
    1. Add an import:  from my_app import views
    2. Add a URL to urlpatterns:  path('', views.home, name='home')
Class-based views
    1. Add an import:  from other_app.views import Home
    2. Add a URL to urlpatterns:  path('', Home.as_view(), name='home')
Including another URLconf
    1. Import the include() function: from django.urls import include, path
    2. Add a URL to urlpatterns:  path('blog/', include('blog.urls'))
"""
from django.urls import path


from django.contrib import admin
from django.conf import settings
from django.conf.urls.static import static
from student.views import studentlogincheck,viewtrainerfilespage
from trainer.models import patientsdata
from trainer.views import trainerlogincheck, uploadfile, uploadtrainingdata,  upload_list
from .views import index, trainer, student, adminlogin, adminloginaction, logout, trainerregistration, \
        viewadmintrainerspage, studentregistration, viewadminstudentpage, activatestudents,
```

```
studentbase, \
    activatetrainers, trainerpage, viewadminfilespage, studentsendquery, viewstudentqueries, \
    adminbase, storecsvdata, viewadmincsvpage, studentpage, chatbot
from django.conf.urls import url
from django.urls import path


from django.urls import path, include


urlpatterns = [
    url(r'^admin/', admin.site.urls),
    url(r'^$',index, name="index"),
    url(r'^index/', index, name="index"),
    url(r'^adminlogin/', adminlogin, name="adminlogin"),
    url(r'^trainer/', trainer, name="trainer"),
    url(r'^student/', student, name="student"),
    url(r'^studentbase/',studentbase, name="studentbase"),
    url(r'^adminbase/',adminbase, name="adminbase"),
    url(r'^adminloginaction/', adminloginaction, name="adminloginaction"),
    url(r'^trainerregistration/', trainerregistration, name="trainerregistration"),
    url(r'^trainerlogincheck/', trainerlogincheck, name="trainerlogincheck"),
    url(r'^trainerpage/', trainerpage, name="trainerpage"),
    url(r'^viewadmintrainerspage/',viewadmintrainerspage,name="viewadmintrainerspage"),
    url(r'^activatetrainers/$', activatetrainers, name="activatetrainers"),
    url(r'^studentlogincheck/', studentlogincheck, name="studentlogincheck"),
    url(r'^studentregistration/', studentregistration, name="studentregistration"),
    url(r'^viewadminstudentpage/',viewadminstudentpage,name="viewadminstudentpage"),
    url(r'^studentpage/',studentpage,name="studentpage"),
    url(r'^activatestudents/$', activatestudents, name="activatestudents"),
    url(r'^uploadfile/',uploadfile,name="uploadfile"),
```

```
    url(r'^uploadtrainingdata/',uploadtrainingdata,name="uploadtrainingdata"),
    url(r'^viewadminfilespage/',viewadminfilespage,name="viewadminfilespage"),
    url(r'^viewtrainerfilespage/',viewtrainerfilespage,name="viewtrainerfilespage"),
    url(r'^storecsvdata/',storecsvdata,name="storecsvdata"),
    url(r'^viewadmincsvpage/',viewadmincsvpage,name="viewadmincsvpage"),
    url(r'^upload_list/',upload_list,name="upload_list"),
    url(r'^studentsendquery',studentsendquery,name="studentsendquery"),
    url(r'^viewstudentqueries/',viewstudentqueries,name="viewstudentqueries"),
    url(r'^chatbot/',chatbot,name="chatbot"),
    url(r'^logout/', logout, name="logout"),

]
if settings.DEBUG:
    urlpatterns += static(settings.MEDIA_URL, document_root=settings.MEDIA_ROOT)
```

Views.py

```
import os
import csv
import json
from collections import defaultdict
from io import TextIOWrapper
from os import name
from tokenize import Name
import requests
from django.http import JsonResponse
from django.conf import settings

from django.contrib import messages
from django.shortcuts import render, HttpResponseRedirect, HttpResponse, redirect
```

```
from random import randint


from django.template.context_processors import request


from MedicalEducation.forms import trainerregistrationform, studentregistrationform, studentqueryform

from MedicalEducation.models import trainerregistrationmodel, studentregistrationmodel, studentquerymodel

from trainer.models import upload, patientdatamodel


import requests

from django.http import JsonResponse

from django.shortcuts import render

from django.http import JsonResponse

import requests

import os


def index(request):

    context = {}

    return render(request, "index.html", context)


def adminbase(request):

    return render(request,'adminbase.html')


def studentbase(request):

    return render(request,'studentbase.html')


def trainerpage(request):

    context = {}

    return  render(request,"trainer/trainerpage.html",context)
```

```
def adminlogin(request):
    return render(request,'adminlogin.html')


def adminloginaction(request):
    if request.method == "POST":
        if request.method == "POST":
            usid = request.POST.get('username')
            pswd = request.POST.get('password')
            if usid == 'admin' and pswd == 'admin':
                return render(request,'admins/adminhome.html')
            else:
                messages.success(request, 'Invalid user id and password')
    #messages.success(request, 'Invalid user id and password')
    return render(request,'adminlogin.html')


def logout(request):
    return render(request,'index.html')


def trainer(request):
    context = {}
    return render(request, "trainer.html", context)


def trainerregistration(request):
    if request.method == 'POST':
        form = trainerregistrationform(request.POST)
        if form.is_valid():
            #print("Hai Meghana")
            form.save()
```

```
            messages.success(request, 'you are successfully registred')
            return HttpResponseRedirect('trainer')
        else:
            print('Invalid')
    else:
        form = trainerregistrationform()
    return render(request,"trainerregistration.html",{'form':form})

def studentregistration(request):
    if request.method == 'POST':
        form = studentregistrationform(request.POST)
        if form.is_valid():
            #print("Hai Meghana")
            form.save()
            messages.success(request, 'you are successfully registred')
            return HttpResponseRedirect('student')
        else:
            print('Invalid')
    else:
        form = studentregistrationform()
    return render(request,"studentregistration.html",{'form':form})

def studentpage(request):
    return render(request, 'studentbase.html')

def studentsendquery(request):
    if request.method == 'POST':
        form = studentqueryform(request.POST)
        if form.is_valid():
```

```
            form.save()
            messages.success(request,'you are successfully send message')
            return HttpResponseRedirect('studentsendquery')
        else:
            print('Invalid')
    else:
        form = studentqueryform()
    return render(request,"studentsendquery.html",{'form':form})

def viewadmintrainerspage(request):
    trainerdata = trainerregistrationmodel.objects.all()
    return render(request,'admins/viewtrainersdata.html',{'object':trainerdata})

def viewadminstudentpage(request):

    studentdata = studentregistrationmodel.objects.all()
    return render(request,'admins/viewstudentsdata.html',{'object':studentdata})

def viewadminfilespage(request):
    filedata = upload.objects.all()
    return render(request,'admins/viewfilesdata.html',{'object':filedata})

def viewadmincsvpage(request):
    csvdata = patientdatamodel.objects.all()
    return render(request,'admins/viewcsvdata.html',{'object':csvdata})


def viewstudentqueries(request):
    sendquery = studentquerymodel.objects.all()
    return render(request,'trainer/viewsendquery.html',{'object':sendquery})
```

```
def activatetrainers(request):
    if request.method=='GET':
        usid = request.GET.get('usid')
        authkey = random_with_N_digits(8)
        status = 'activated'
        print("USID = ",usid,authkey,status)
        trainerregistrationmodel.objects.filter(id=usid).update(authkey=authkey , status=status)
        trainerdata = trainerregistrationmodel.objects.all()
        return render(request,'admins/viewtrainersdata.html',{'object':trainerdata})

def activatestudents(request):
    if request.method=='GET':
        pid = request.GET.get('pid')
        authkey = random_with_N_digits(8)
        status = 'activated'
        print("PID = ",pid,authkey,status)
        studentregistrationmodel.objects.filter(id=pid).update(authkey=authkey , status=status)
        studentdata = studentregistrationmodel.objects.all()
        return render(request,'admins/viewstudentsdata.html',{'object':studentdata})



def random_with_N_digits(n):
    range_start = 10**(n-1)
    range_end = (10**n)-1
    return randint(range_start, range_end)

def student(request):
    context = {}
```

```
    return render(request, "student.html", context)


def storecsvdata(request):
    if request.method == 'POST':
        name = request.POST.get('name')
        csvfile =TextIOWrapper( request.FILES['file'])
        columns = defaultdict(list)
        #with open('trainer/static/upload/patientdata.csv') as csvfile:
            #storecsvdata = csv.reader(csvfile)
            #row1 = next(storecsvdata)
            #row2 = next(storecsvdata)
            #print(row1)
            #print(row2)
        storecsvdata = csv.DictReader(csvfile)
            #row1 = next(storecsvdata)
        for row1 in storecsvdata:
                name = row1["Name"]
                email = row1["Email"]
                dob = row1["DOB"]
                address = row1["Address"]
                pan = row1["PAN"]
                city = row1["City"]
                country = row1["Country"]
                patientdatamodel.objects.create(Name=name, Email=email, DOB=dob,
                                                Address=address, PAN=pan, City=city,
                                                Country=country)
                #for (k, v) in row1.items():
                    #var = columns[k].append(v)
                    #print("Values ",var," k  value ",k," V value ",v)
```

```
            #patientdatamodel.objects.create(Name=columns['Name'], Email=columns['Email'], DOB=columns['DOB'], Address=columns['Address'], PAN=columns['PAN'], City=columns['City'], Country=columns['Country'], )


      #patientdatamodel.objects.create(Name=columns['Name'],Email=columns['Email'],DOB=columns['DOB'],Address=columns['Address'],PAN=columns['PAN'],City=columns['City'],Country=columns['Country'],)

    #print(columns['Name'])

    #print(columns['Email'])

    #print(columns['DOB'])


    print("Name is ",csvfile)

    return HttpResponse('CSV file successful uploaded')

  else:


    return render(request, 'patientsdata.html', {})


def chatbot(request):

  return render(request, 'chatbot/chatbot.html')
```

forms.py

```
import os
from uuid import uuid4

from django import forms
from django.forms import HiddenInput

from MedicalEducation.models import trainerregistrationmodel, studentregistrationmodel, studentquerymodel
from trainer.models import upload, patientsdata

class trainerregistrationform(forms.ModelForm):
  name = forms.CharField(widget=forms.TextInput(), required=True, max_length=100)
  loginid = forms.CharField(widget=forms.TextInput(), required=True, max_length=100)
  password = forms.CharField(widget=forms.PasswordInput(), required=True, max_length=100)
  email = forms.EmailField(widget=forms.TextInput(),required=True)
```

```
    qualification = forms.CharField(widget=forms.TextInput(),required=True,max_length=100)
    mobile = forms.CharField(widget=forms.TextInput(),required=True,max_length=100)
    address = forms.CharField(widget=forms.TextInput(),required=True,max_length=100)
    state = forms.CharField(widget=forms.TextInput(),required=True,max_length=100)
    authkey = forms.CharField(widget=forms.HiddenInput(), initial='waiting', max_length=100)
    status = forms.CharField(widget=forms.HiddenInput(), initial='waiting', max_length=100)

    class Meta:
        model = trainerregistrationmodel
        fields =
['name','loginid','password','email','qualification','mobile','address','state','authkey','status' ]

class studentregistrationform(forms.ModelForm):
    name = forms.CharField(widget=forms.TextInput(), required=True, max_length=100)
    loginid = forms.CharField(widget=forms.TextInput(), required=True, max_length=100)
    password = forms.CharField(widget=forms.PasswordInput(), required=True,
max_length=100)
    email = forms.CharField(widget=forms.TextInput(), required=True)
    college = forms.CharField(widget=forms.TextInput(), required=True, max_length=100)
    mobile = forms.CharField(widget=forms.TextInput(), required=True, max_length=100)
    address = forms.CharField(widget=forms.TextInput(), required=True, max_length=100)
    state = forms.CharField(widget=forms.TextInput(), required=True, max_length=100)
    authkey = forms.CharField(widget=forms.HiddenInput(), initial='waiting', max_length=100)
    status = forms.CharField(widget=forms.HiddenInput(), initial='waiting', max_length=100)

    class Meta:
        model = studentregistrationmodel
        fields = ['name','loginid','password','email','college','mobile','address', 'state',
'authkey','status']


class studentqueryform(forms.ModelForm):
    name = forms.CharField(widget=forms.TextInput(), required=True, max_length=100)
    email = forms.CharField(widget=forms.TextInput(), required=True)
    query = forms.CharField(widget=forms.TextInput(), required=True, max_length=500)

    class Meta:
        model = studentquerymodel
        fields = ['name', 'email', 'query']


class UploadfileForm(forms.ModelForm):
    #filename = forms.CharField(widget=forms.TextInput(),required=True, max_length=100)
    #description = forms.CharField(widget=forms.TextInput(),required=True,max_length=100)
    #file = forms.FileField()
    class Meta:
        model = upload
        fields = ('filename','description','file')

"""class patientsdataForm(forms.ModelForm):
```

```
    class Meta:
        model = patientsdata
        fields = ('filename','description','file')"""
```

models.py

```
from django.db import models

class trainerregistrationmodel(models.Model):
    name = models.CharField(max_length=100)
    loginid = models.CharField(max_length=100)
    password = models.CharField(max_length=100)
    email = models.EmailField()
    qualification = models.CharField(max_length=100)
    mobile = models.CharField(max_length=100)
    address = models.CharField(max_length=100)
    state = models.CharField(max_length=100)
    authkey = models.CharField(max_length=100)
    status = models.CharField(max_length=100)

def __str__(self):
    return self.email

class studentregistrationmodel(models.Model):
    name = models.CharField(max_length=100)
    loginid = models.CharField(max_length=100)
    password = models.CharField(max_length=100)
    email = models.EmailField()
    college = models.CharField(max_length=100)
    mobile = models.CharField(max_length=100)
    address = models.CharField(max_length=100)
    state = models.CharField(max_length=100)
    authkey = models.CharField(max_length=100)
    status  = models.CharField(max_length=100)

    def __str__(self):
        return self.email

class studentquerymodel(models.Model):
    name = models.CharField(max_length=100)
    email = models.CharField(max_length=100)
    query = models.CharField(max_length=500)

    def __str__(self):
        return self.emailid
```

Trainer’s Views.py

```
import csv,os
```

```
import io
from msilib.schema import File, ListView
from pathlib import Path
from uuid import uuid4

from django.contrib import messages, auth
from django.contrib.auth.decorators import permission_required
from django.http import HttpResponse, HttpResponseRedirect
from django.shortcuts import render, redirect

import trainer
from MedicalEducation import settings
from MedicalEducation.forms import UploadfileForm
from MedicalEducation.models import trainerregistrationmodel
from trainer.functions.functions import handle_uploaded_file

def trainerlogincheck(request):
    if request.method == 'POST':
        usid = request.POST.get('loginid')
        #print(usid)
        pswd = request.POST.get('password')
        #print(pswd)
        try:
            check = trainerregistrationmodel.objects.get(loginid=usid, password=pswd)
           # print('usid',usid,'pswd',pswd)
            request.session['traid'] = check.loginid
            request.session['loggeduser'] = check.name
            status = check.status
            #print("stu  id ", check.id)
            #if usid=='loginid' and pswd=='loggeduser':
            #if usid is not None:
            if status == "activated":
                request.session['email'] = check.email
                #auth.login(request, usid)
                return render(request,'trainer/trainerpage.html')
            else:
                messages.success(request, 'trainer is not activated')
                return render(request,'trainer.html')

            return render(request,'trainer/trainerpage.html')
        except Exception as e:
            print('Exception is ', str(e))
            messages.success(request,'Invalid user id and password')
        return render(request,'trainer.html')

"""def uploadfile(request):
    #print("Am Arumalla")
    if request.method == 'POST':
        #print("Was Up")
        upload = UploadfileForm(request.POST, request.FILES)
```

```
        if upload.is_valid():
            #handle_uploaded_file(request.FILES['file'])
            upload.save()
            return HttpResponse("File Upload Sucessfully")
        else:
            upload = UploadfileForm()
            return render(request, 'uploadfile.html', {'form': upload})
    else:
        upload = UploadfileForm()
        return render(request,'uploadfile.html',{'form':upload})

    #return render(request, 'uploadfile.html', {})
    #return HttpResponseRedirect("uploadtrainingdata")"""

def uploadfile(request):
    if request.method == 'POST':
        form = UploadfileForm(request.POST, request.FILES)
        if form.is_valid():
            form.save()
            return redirect('upload_list.html')
    else:
        form = UploadfileForm()
    return render(request, 'uploadfile.html', {'form': form})

def upload_list(request):
    files = File.objects.all()
    return render(request, 'upload_list.html', {'files': files})

"""def delete_file(request, pk):
    if request.method == 'POST':
        file = File.objects.get(pk=pk)
        file.delete()
    return redirect('upload_list')"""

def uploadtrainingdata(request):
    return render(request,"uploadfile.html",{})


"""@permission_required('admin.can_add_log_entry')
def address_upload(request):
    template = "address_upload.html"

    prompt = {
        'order': 'Order of the csv should be name, email, address, healthcondition'
    }

    if request.method == "GET":
        return  render(request,template,prompt)
    csv_file = request.FILES['file']
```

```
    if not csv_file.name.endswith('.csv'):
        messages.error(request,'This is not a csv file')

    data_set = csv_file.read().decode('UTF-8')
    io_string = io.StringIO(data_set)
    next(io_string)
    for column in csv.reader(io_string,delimiter=',',quotechar='|'):
        _, created = trainer.objects.update_or_create(
            name=column[0],
            email=column[1],
            address=column[2],
            healthcondition=column[3]
        )
    context = {}
    return render(request, template, context)"""
```

Student's Views.py

```
from django.contrib import messages
from django.shortcuts import render

from MedicalEducation.models import studentregistrationmodel
from trainer.models import upload

def studentlogincheck(request):
    if request.method == "POST":
        usid = request.POST.get('loginid')
        pswd = request.POST.get('password')
        try:
            check = studentregistrationmodel.objects.get(loginid=usid, password=pswd)
            request.session['stuid'] = check.loginid
            request.session['loggedstu'] = check.name
            status = check.status
            print("stu  id ",check.loginid)
            if status == "activated":
               #request.session['stuid'] = check.loginid
               request.session['email'] = check.email
               return render(request,'student/studentpage.html')
            else:
                messages.success(request, 'Your Account Not at activated')
                return render(request,'student.html')

            return render(request,'student/studentpage.html')
        except Exception as e:
            print('Exception is ',str(e))
    messages.success(request, 'Invalid Login Details')
    return render(request,'student.html')

def viewtrainerfilespage(request):
    filedata = upload.objects.all()
```

```
    return render(request,'student/viewtrainerfilesdata.html',{'object':filedata})
```

Adminbase.html

```
{% load static %}
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Admin - MediLearn</title>
    <meta name="description" content="Admin Dashboard" />
    <meta name="keywords" content="AI, Medical, Education" />
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <!-- Bootstrap 5 (Only One Version) -->
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">

    <!-- FontAwesome (for icons) -->
    <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/6.0.0/css/all.min.css">

    <!-- Custom Styles -->
    <style>
        /* Sidebar Styling */
        .sidebar {
            width: 250px;
            height: 100vh;
            position: fixed;
            background: rgb(24, 56, 145);
            color: white;
            padding: 20px;
        }
        .sidebar a {
            color: white;
            text-decoration: none;
            display: block;
            padding: 10px;
            margin: 10px 0;
            border-radius: 5px;
            transition: 0.3s;
        }
        .sidebar a:hover {
            background: rgba(255, 255, 255, 0.2);
        }
        .sidebar a.active {
            background: #007bff;
            font-weight: bold;
        }
        .content {
```

```
            margin-left: 270px;
            padding: 20px;
        }
        .navbar {
            background: #0056b3;
            color: white;
            padding: 10px;
        }
        .card {
            border-radius: 10px;
            box-shadow: 2px 2px 10px rgba(0, 0, 0, 0.1);
        }
        .dashboard {
            display: flex;
            justify-content: space-around;
            text-align: center;
            margin-top: 30px;
        }
        .dashboard-card {
            background: white;
            padding: 20px;
            border-radius: 10px;
            box-shadow: 0 4px 12px rgba(0, 0, 0, 0.2);
            width: 30%;
        }
        .dashboard-card h3 {
            margin: 10px 0;
            font-size: 20px;
            color: #333;
        }
        .count {
            font-size: 28px;
            font-weight: bold;
            color: #007bff;
        }
    </style>
</head>
<body>

<!-- Sidebar -->
<div class="sidebar">
    <h2><b>Admin Dashboard</b></h2>

    <a href="{% url 'adminbase' %}" class="{% if request.resolver_match.url_name ==
'adminbase' %}active{% endif %}">
        <i class="fas fa-home"></i> Admin Home
    </a>

    <a href="{% url 'viewadminstudentpage' %}" class="{% if request.resolver_match.url_name
== 'viewadminstudentpage' %}active{% endif %}">
```

```
      <i class="fas fa-user-graduate"></i> Student Details
    </a>

    <a href="{% url 'viewadmintrainerspage' %}" class="{% if request.resolver_match.url_name
== 'viewadmintrainerspage' %}active{% endif %}">
      <i class="fas fa-user-tie"></i> Trainer Details
    </a>

    <a href="{% url 'viewadminfilespage' %}" class="{% if request.resolver_match.url_name ==
'viewadminfilespage' %}active{% endif %}">
      <i class="fas fa-file"></i> File Details
    </a>

    <a href="{% url 'logout' %}">
      <i class="fas fa-sign-out-alt"></i> Logout
    </a>
</div>

<!-- Main Content Wrapper -->
<div class="content">
    <!-- Top Navbar -->
    <nav class="navbar navbar-expand-lg navbar-dark">
      <div class="container-fluid">
        <span class="navbar-brand">Admin Panel</span>
        <div class="ms-auto">
          <span class="me-3">Welcome, Admin</span>
          <a href="{% url 'logout' %}" class="btn btn-danger btn-sm">Logout</a>
        </div>
      </div>
    </nav>

    <!-- Page Content Block -->
    {% block contents %}
    <div class="container mt-4">
      <h2 class="text-primary text-center">📊 Admin Dashboard</h2>

      <div class="dashboard">
        <div class="dashboard-card">
          <h3>🧑‍🏫 Trainers Registered</h3>
          <div class="count">{{ trainer_count }}</div>
        </div>

        <div class="dashboard-card">
          <h3>🎓 Students Registered</h3>
          <div class="count">{{ student_count }}</div>
        </div>

        <div class="dashboard-card">
          <h3>📂 Uploaded Files</h3>
          <div class="count">{{ file_count }}</div>
```

```
        </div>
      </div>
    </div>
    {% endblock %}
</div>

</body>
</html>
```

Trainerbase.html

```
{% load static %}
<!DOCTYPE html>
<html lang="en">

<head>
    <title>Trainer - MediLearn</title>
    <meta name="description" content="AI in Medical Education" />
    <meta name="keywords" content="AI, Medical, Education, Training" />
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <!-- Bootstrap CSS -->
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">

    <!-- Custom CSS -->
    <link rel="stylesheet" href="{% static 'css/styler.css' %}">

    <style>
        /* Fullscreen Loader */
        #loader {
            position: fixed;
            width: 100%;
            height: 100%;
            background: url("{% static 'image/background3.jpg' %}") no-repeat center center/cover;
            background-attachment: fixed;
            z-index: 9999;
            display: flex;
            justify-content: center;
            align-items: center;
            transition: opacity 0.5s ease-in-out;
        }

        /* Loader Overlay - Black Transparency */
        #loader::after {
            content: "";
            position: absolute;
            top: 0;
            left: 0;
```

```
    width: 100%;
    height: 100%;
    background: rgba(0, 0, 0, 0.4);
    z-index: -1;
}

/* Navbar */
.navbar {
    background: rgba(0, 0, 0, 0.7) !important;
    position: fixed;
    width: 100%;
    top: 0;
    left: 0;
    z-index: 9999;
}

/* Adjust Body to Prevent Content Overlapping Navbar */
body {
    padding-top: 70px;
    background: url("{% static 'image/background3.jpg' %}") no-repeat center center/cover;
    background-attachment: fixed;
    position: relative;
}

body::after {
    content: "";
    position: fixed;
    top: 0;
    left: 0;
    width: 100%;
    height: 100%;
    background: rgba(0, 0, 0, 0.4);
    z-index: -1;
}

/* Navbar Brand Logo */
.navbar-brand img {
    height: 50px;
}

/* Navbar Links */
.navbar-nav .nav-link {
    color: white !important;
    position: relative;
}

.navbar-nav .nav-link::after {
    content: "";
    position: absolute;
    left: 0;
```

```
            bottom: -2px;
            width: 100%;
            height: 2px;
            background-color: #007bff;
            transform: scaleX(0);
            transition: transform 0.3s ease-in-out;
        }

        .navbar-nav .nav-link:hover::after,
        .navbar-nav .nav-link.active::after {
            transform: scaleX(0.85);
        }

        /* Footer */
        footer {
            background: #f8f9fa;
            padding: 15px 0;
        }
    </style>
</head>

<body>

    <!-- Page Loader -->
    <div id="loader">
        <div class="spinner-border text-primary" role="status">
            <span class="visually-hidden">Loading...</span>
        </div>
    </div>

    <!-- Navbar -->
    <nav class="navbar navbar-expand-lg navbar-dark bg-dark">
        <div class="container">
            <!-- Logo -->
            <a class="navbar-brand" href="{% url 'trainerpage' %}">
                <img src="{% static 'image/logo.jpg' %}" alt=" MediLearn Academy" height="50">
            </a>

            <!-- Navbar Toggle Button -->
            <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-
target="#navbarNav">
                <span class="navbar-toggler-icon"></span>
            </button>

            <!-- Navbar Links -->
            <div class="collapse navbar-collapse" id="navbarNav">
                <ul class="navbar-nav ms-auto">
                    <li class="nav-item">
                        <a class="nav-link {% if request.resolver_match.url_name == 'trainerpage'
%}active{% endif %}" href="{% url 'trainerpage' %}">Home</a>
```

```
                    </li>
                    <li class="nav-item">
                        <a class="nav-link {% if request.resolver_match.url_name ==
'viewstudentqueries' %}active{% endif %}" href="{% url 'viewstudentqueries' %}">Queries</a>
                    </li>
                    <li class="nav-item">
                        <a class="nav-link {% if request.resolver_match.url_name == 'uploadfile'
%}active{% endif %}" href="{% url 'uploadfile' %}">Upload Materials</a>
                    </li>
                    <li class="nav-item">
                        <a class="nav-link" href="{% url 'logout' %}">Logout</a>
                    </li>
                </ul>
            </div>
        </div>
    </nav>

    <!-- Main Content -->
    <div class="container">
        {% block contents %} {% endblock %}
    </div>
    {% block content %}{% endblock %}

    <!-- JavaScript to Hide Loader After Page Load -->
    <script>
        window.onload = function() {
            setTimeout(() => {
                document.getElementById("loader").style.opacity = "0";
                setTimeout(() => document.getElementById("loader").style.display = "none", 500);
            }, 800);
        };
    </script>

    <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
</body>

</html>
```

Studentbase.html

```
{% load static %}

<!doctype html>
<html lang="en">

<head>
  <title>Student - MediLearn </title>
  <meta name="description" content="website description" />
  <meta name="keywords" content="website keywords, website keywords" />
```

```
  <meta http-equiv="content-type" content="text/html; charset=UTF-8" />
  <link rel="stylesheet" href="{% static 'css/styler.css' %}">
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">

<style>
/* Fullscreen Loader */
#loader {
    position: fixed;
    width: 100%;
    height: 100%;
    background: url("{% static 'image/background2.jpg' %}") no-repeat center center/cover;
    background-attachment: fixed;
    z-index: 9999;
    display: flex;
    justify-content: center;
    align-items: center;
    transition: opacity 0.5s ease-in-out;
}

/* Loader Overlay - Black Transparency */
#loader::after {
    content: "";
    position: absolute;
    top: 0;
    left: 0;
    width: 100%;
    height: 100%;
    background: rgba(0, 0, 0, 0.5); /* Adjust transparency */
    z-index: -1;
}

/* Navbar */
.navbar {
    background: rgba(0, 0, 0, 0.8) !important;
    position: fixed;
    width: 100%;
    top: 0;
    left: 0;
    z-index: 9999;
    margin-top: -45px;
}

/* Body Background with Black Transparency */
body {
    padding-top: 70px; /* Prevent content from hiding behind navbar */
    background: url("{% static 'image/background2.jpg' %}") no-repeat center center/cover;
    background-attachment: fixed;
    position: relative;
}
```

```css
/* Black Transparent Overlay */
body::after {
    content: "";
    position: fixed;
    top: 0;
    left: 0;
    width: 100%;
    height: 100%;
    background: rgba(0, 0, 0, 0.65); /* Adjust transparency */
    z-index: -1;
}

.navbar-brand img {
    height: 50px;
}

.navbar-nav .nav-link {
    color: white !important;
    position: relative;
}

.navbar-nav .nav-link::after {
    content: "";
    position: absolute;
    left: 0;
    bottom: -2px;
    width: 100%;
    height: 2px;
    background-color: #007bff;
    transform: scaleX(0);
    transition: transform 0.3s ease-in-out;
}

.navbar-nav .nav-link:hover::after,
.navbar-nav .nav-link.active::after {
    transform: scaleX(0.85);
}

.container {
    margin-top: 50px;
}

.course-container {
    display: flex;
    justify-content: center;
    gap: 20px;
    flex-wrap: wrap;
    margin-top: 30px;
```

```
}

.course-card {
   width: 220px;
   text-align: center;
   background: white;
   padding: 15px;
   border-radius: 10px;
   box-shadow: 0 4px 8px rgba(28, 20, 20, 0.2);
   transition: transform 0.3s ease-in-out;
}

.course-card:hover {
   transform: scale(1.05);
}

.course-card img {
   width: 100%;
   border-radius: 5px;
}

.course-title {
   position: relative;
   display: inline-block;
   padding-bottom: 10px; /* Space between text and underline */
}
/* Remove default link styling */
.course-card {
   text-decoration: none;
   color: black;
   transition: transform 0.3s ease, box-shadow 0.3s ease;
}

/* Hover effect */
.course-card:hover {
   transform: scale(1.05);
   box-shadow: 0 5px 15px rgba(0, 123, 255, 0.2);
}

/* Improved course title underline */
.course-title {
   position: relative;
   display: inline-block;
   padding-bottom: 10px; /* Space between text and underline */
}

.course-title::after {
   content: "";
   position: absolute;
   left: 50%;
```

```
    bottom: 0;
    width: 80px; /* Adjust width */
    height: 3px; /* Thickness */
    background-color: #007bff;
    transform: translateX(-50%);
}

.course-title::after {
    content: "";
    position: absolute;
    left: 50%;
    bottom: 0;
    width: 50%; /* Adjust width of the underline */
    height: 3px; /* Thickness of the underline */
    background-color: #007bff;
    transform: translateX(-50%);
}

footer {
    background: #f8f9fa;
    padding: 15px 0;
}

  </style>
</head>

<body>
     <!-- Page Loader -->
  <div id="loader">
    <div class="spinner-border text-primary" role="status">
      <span class="visually-hidden">Loading...</span>
    </div>
  </div>

  <!-- Navbar -->
  <nav class="navbar navbar-expand-lg navbar-dark bg-dark">
    <div class="container">
      <a class="navbar-brand" href="{% url 'studentbase' %}">
        <img src="{% static 'image/logo.jpg' %}" alt="Technova Academy" height="50">
      </a>
      <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-
target="#navbarNav">
        <span class="navbar-toggler-icon"></span>
      </button>
      <div class="collapse navbar-collapse" id="navbarNav">
        <ul class="navbar-nav ms-auto">
          <li class="nav-item">
            <a class="nav-link {% if request.resolver_match.url_name == 'studentpage' %}active{%
endif %}" href="{% url 'studentpage' %}">Home</a>
          </li>
```

```
        <li class="nav-item">
          <a class="nav-link {% if request.resolver_match.url_name == 'viewtrainerfilespage' %}active{% endif %}" href="{% url 'viewtrainerfilespage' %}">Materials</a>
        </li>
        <li class="nav-item">
          <a class="nav-link {% if request.resolver_match.url_name == 'studentsendquery' %}active{% endif %}" href="{% url 'studentsendquery' %}">Send Queries</a>
        </li>

        <li class="nav-item">
          <a class="nav-link {% if request.resolver_match.url_name == 'chatbot' %}active{% endif %}" href="{% url 'chatbot' %}">Chat with AI</a>
        </li>
        <li class="nav-item">
          <a class="nav-link" href="{% url 'logout' %}">Logout</a>
        </li>
      </ul>
    </div>
  </div>
</nav>

<div class="container">
  {% block contents %} {% endblock %}
</div>
{% block content %}{% endblock %}


  </div>
  <!-- JavaScript to Hide Loader After Page Load -->
  <script>
      window.onload = function() {
          setTimeout(() => {
              document.getElementById("loader").style.opacity = "0";
              setTimeout(() => document.getElementById("loader").style.display = "none", 500);
          }, 800);
      };
    </script>

<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

6. SYSTEM TEST

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub assemblies, assemblies and/or a finished product It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

6.1 TYPES OF TESTS

6.1.1 Unit testing

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application .it is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

6.1.2 Integration testing

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfaction, as shown by successfully unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

6.1.3 Functional test

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Functional testing is centered on the following items:

Valid Input : identified classes of valid input must be accepted.

Invalid Input : identified classes of invalid input must be rejected.

Functions : identified functions must be exercised.

Output : identified classes of application outputs must be exercised.

Systems/Procedures : interfacing systems or procedures must be invoked.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

6.1.4 System Test

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

6.1.5 White Box Testing

White Box Testing is a testing in which in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose. It is purpose. It is used to test areas that cannot be reached from a black box level.

6.1.6 Black Box Testing

Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most other kinds of tests, must be written from a definitive source document, such as specification or requirements document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box . You cannot "see" into it. The test provides inputs and responds to outputs without considering how the software works.

6.1.7 Unit Testing

Unit testing is usually conducted as part of a combined code and unit test phase of the software lifecycle, although it is not uncommon for coding and unit testing to be conducted as two distinct phases.

6.1.8 Test strategy and approach

Field testing will be performed manually and functional tests will be written in detail.

Test objectives

- All field entries must work properly.
- Pages must be activated from the identified link.
- The entry screen, messages and responses must not be delayed.

Features to be tested

- Verify that the entries are of the correct format

- No duplicate entries should be allowed
- All links should take the user to the correct page.

6.1.9 Acceptance Testing

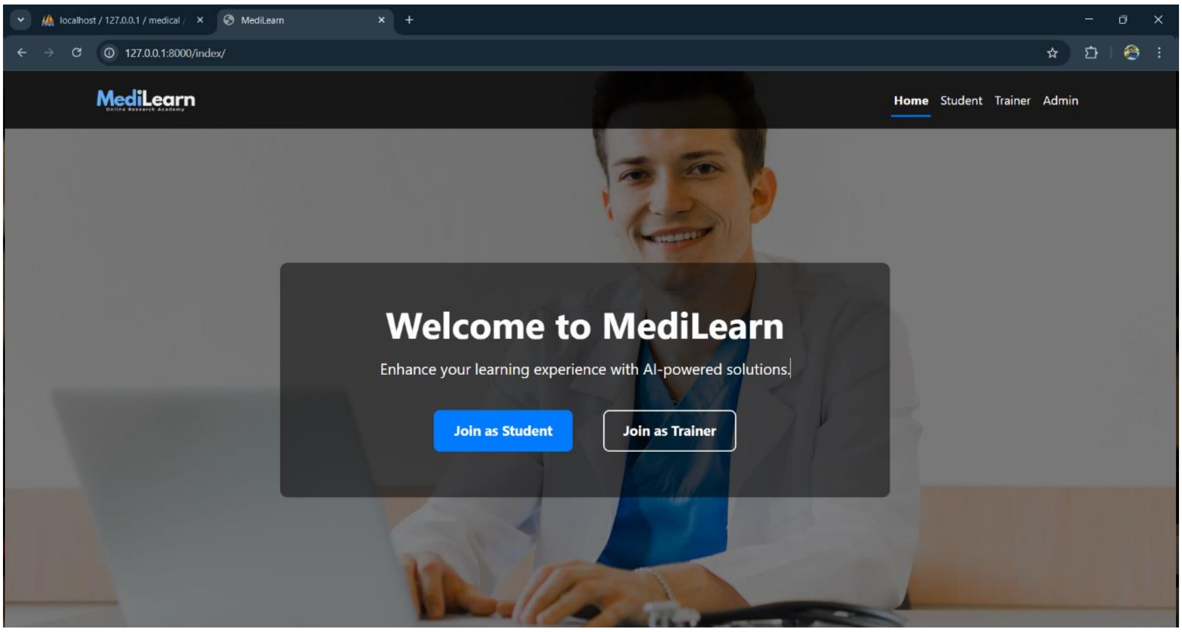
User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

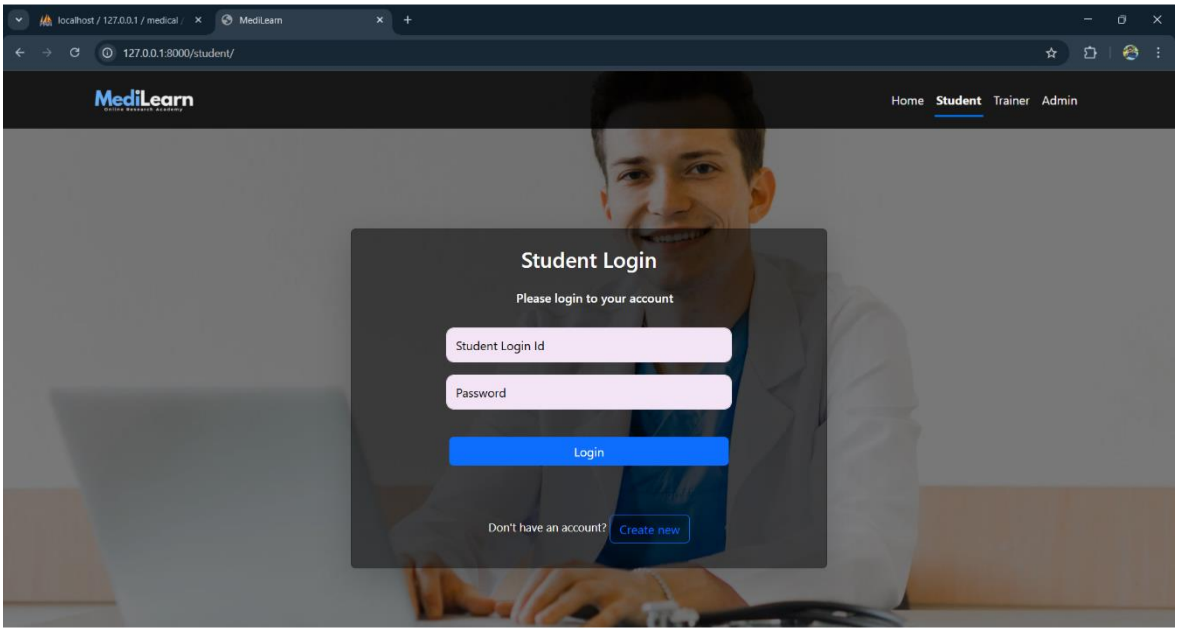
7. RESULTS

MAIN MODULE

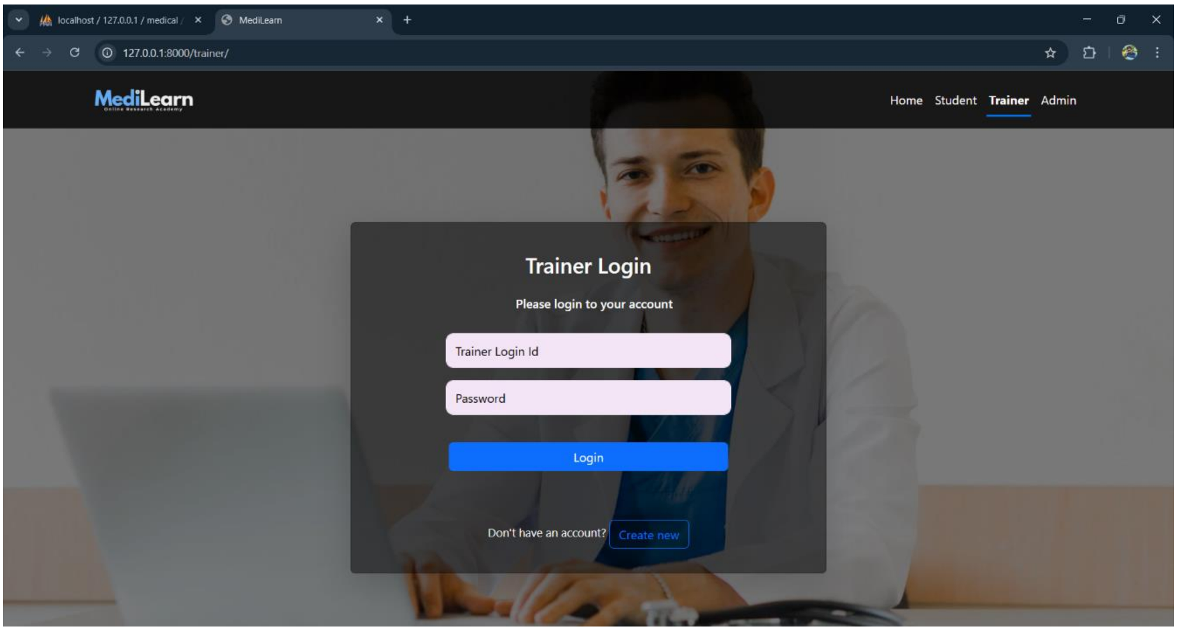
Home (Get Started): The home page serves as the entry point for users, providing an overview of the platform and its features. It guides students, trainers, and admins to their respective login and registration sections.



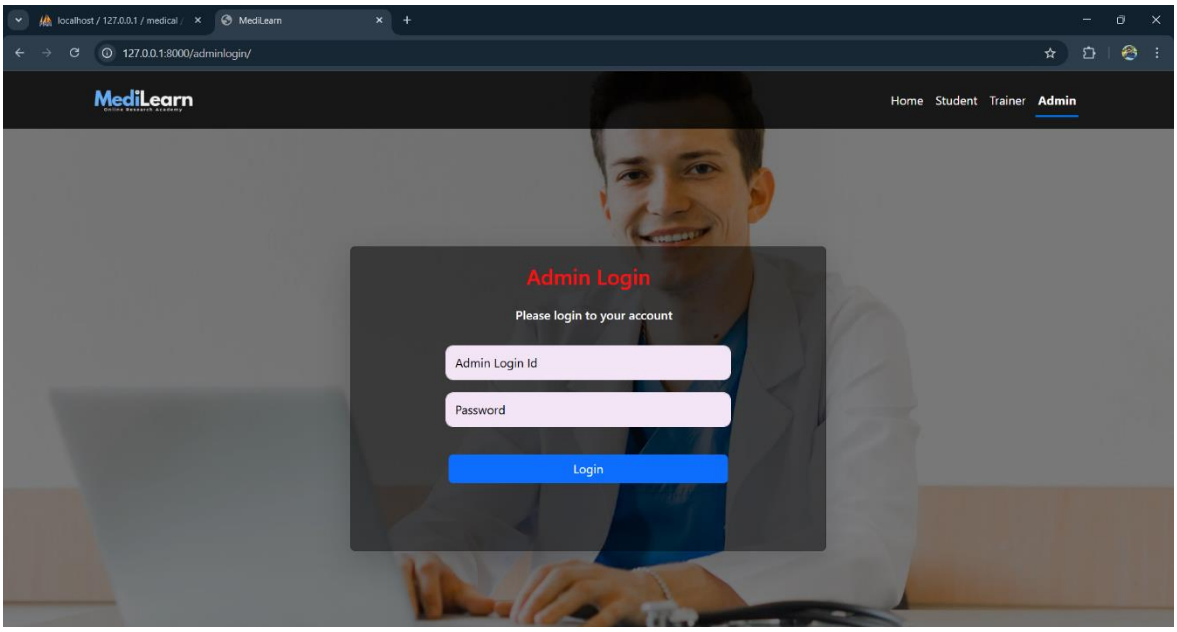
Student Login and Registration: Students can register by providing necessary details and gain access to learning materials. The login feature ensures secure access to their personalized dashboard for study resources and AI-based assistance.



Trainer Login and Registration: Trainers can sign up to manage course materials and interact with students. Once registered, they can log in to upload content, respond to queries, and track student engagement.



Admin Login: Admins log in to oversee the platform's operations, manage user registrations, and ensure system security. They monitor student and trainer activities to maintain smooth functioning of the LMS**.**



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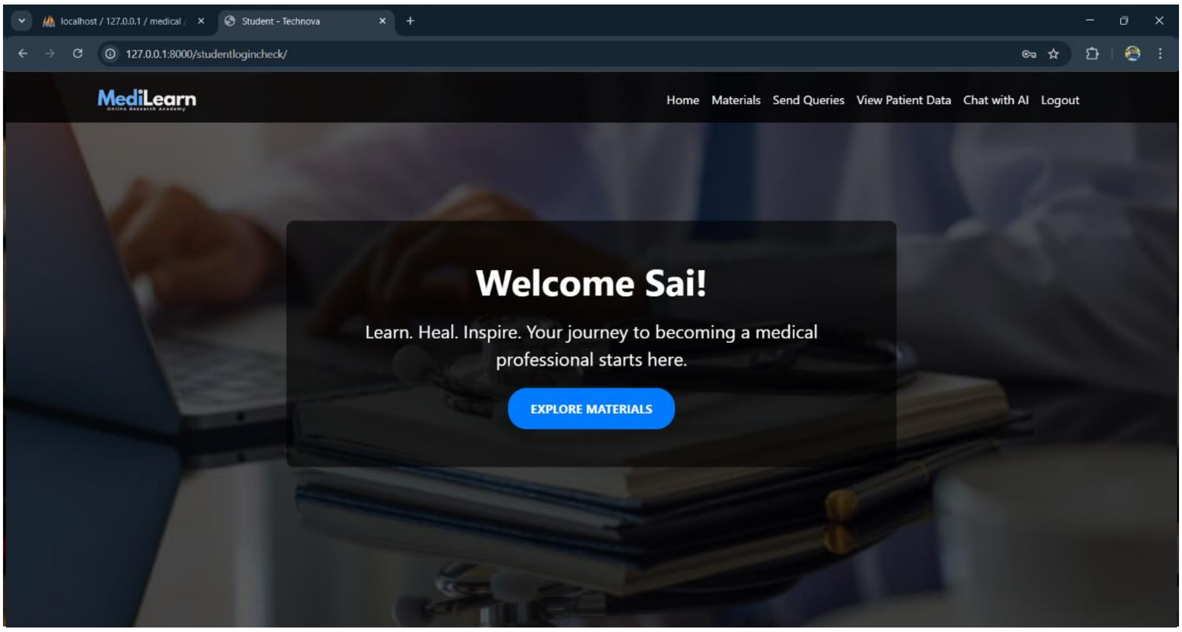
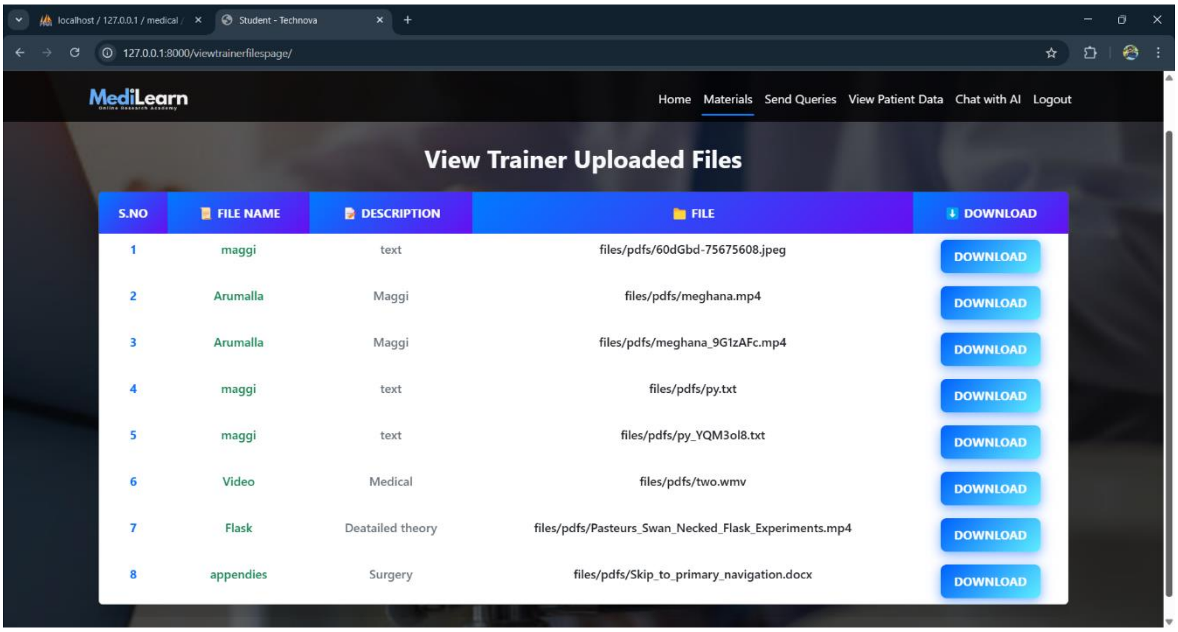
STUDENT MODULE

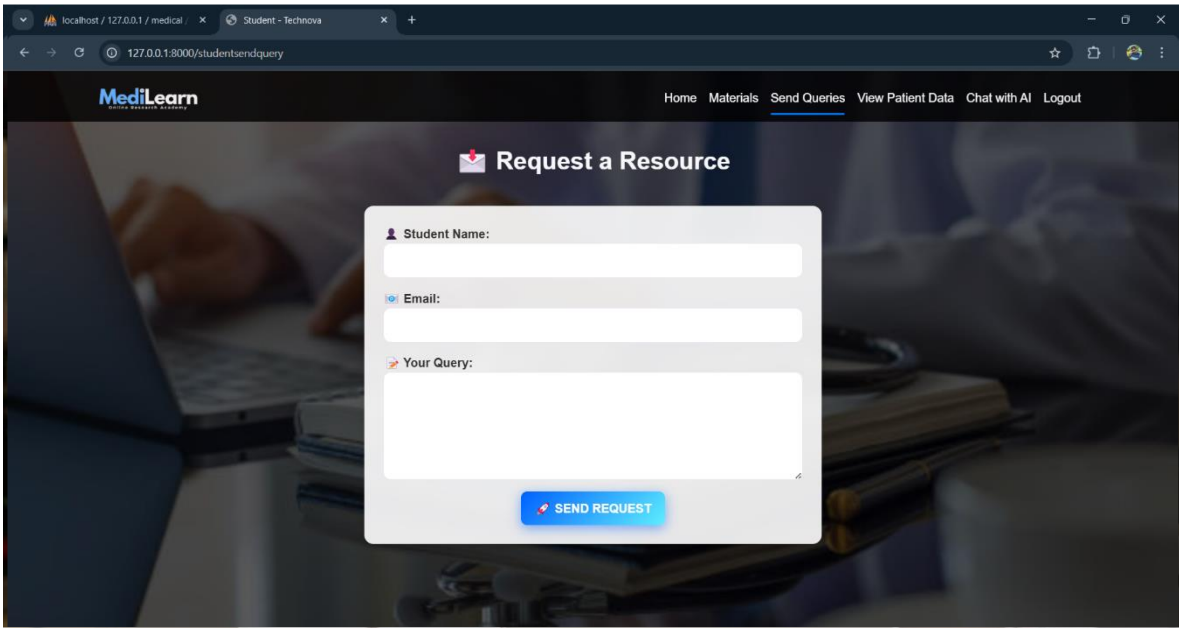
Student Home: The student home page provides an intuitive dashboard with quick access to learning materials and query features. It serves as a centralized space for students to navigate their academic resources efficiently.



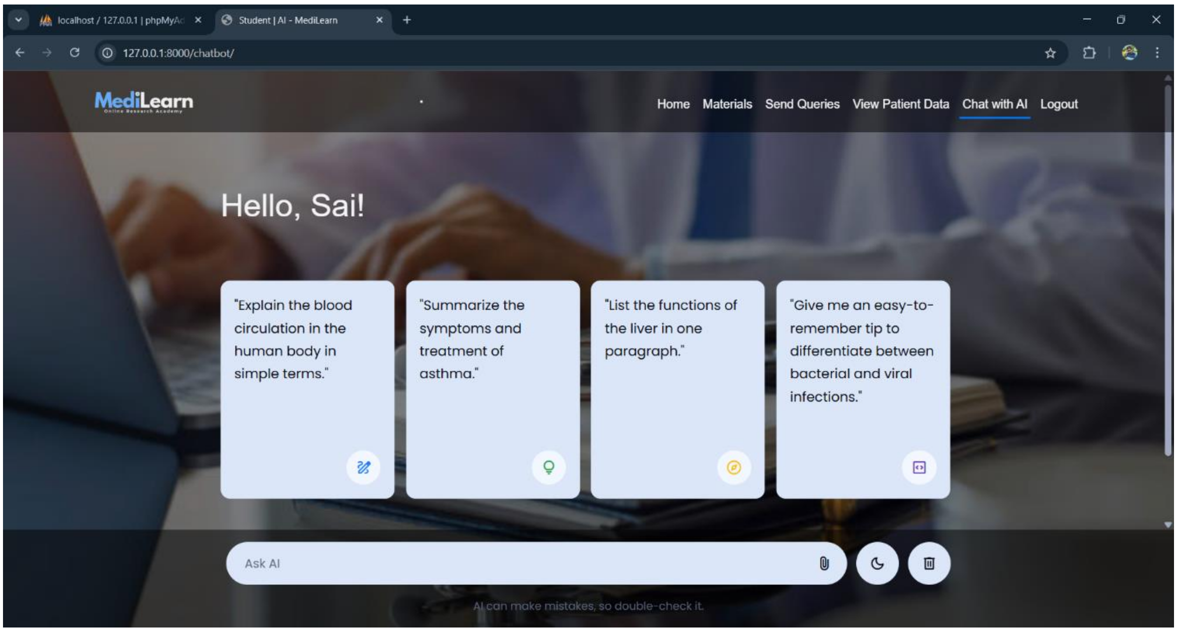
Materials: Students can access course materials uploaded by trainers, ensuring they have the necessary resources for learning. These materials include lecture notes, reference documents, and multimedia content for better understanding.



Send Queries: Students can send their doubts and questions directly to trainers through the platform. This feature enables structured communication, ensuring timely responses and better clarification of concepts.

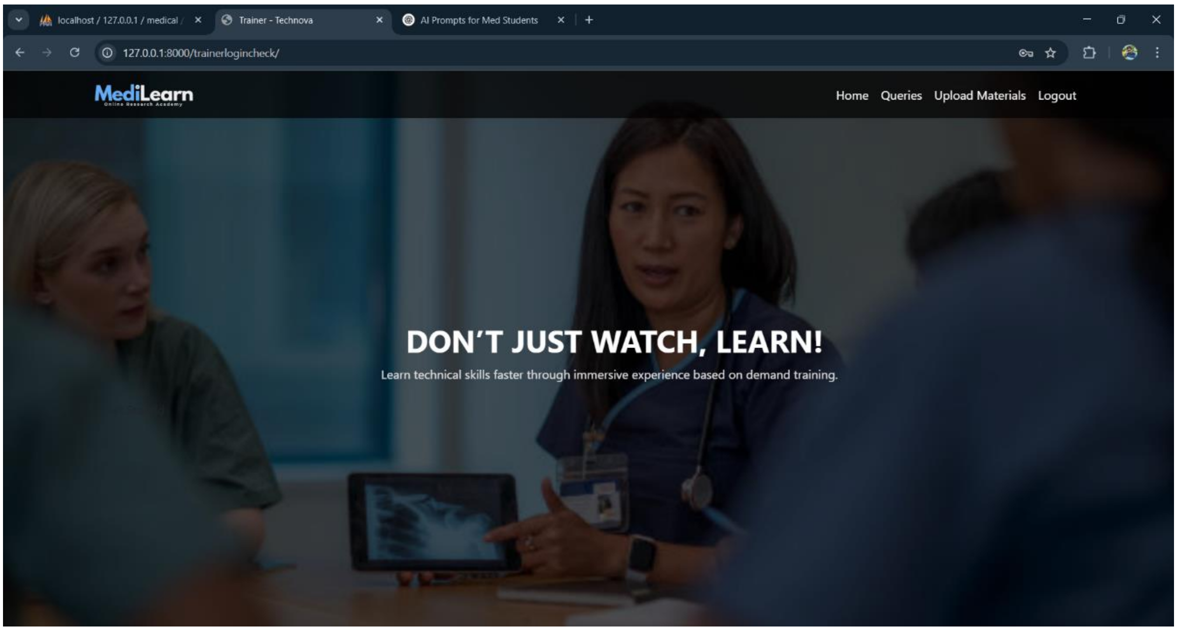


Chat with AI: An AI-powered chatbot assists students by answering queries instantly. It provides explanations, study tips, and guidance, enhancing the learning experience through interactive support.

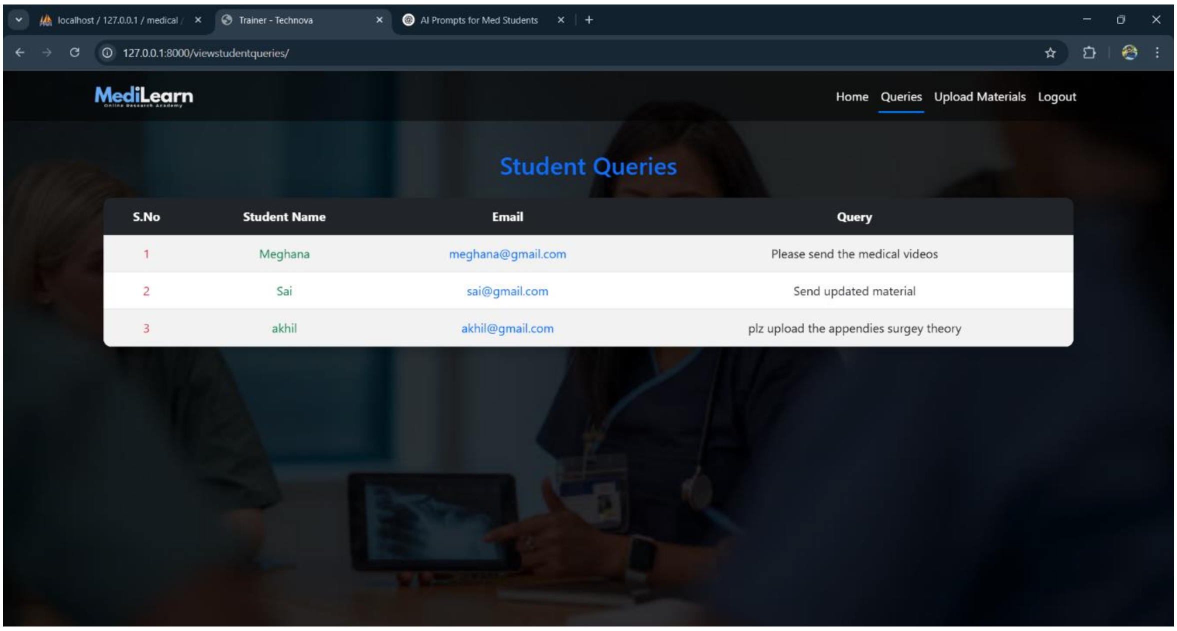


TRAINER MODULE

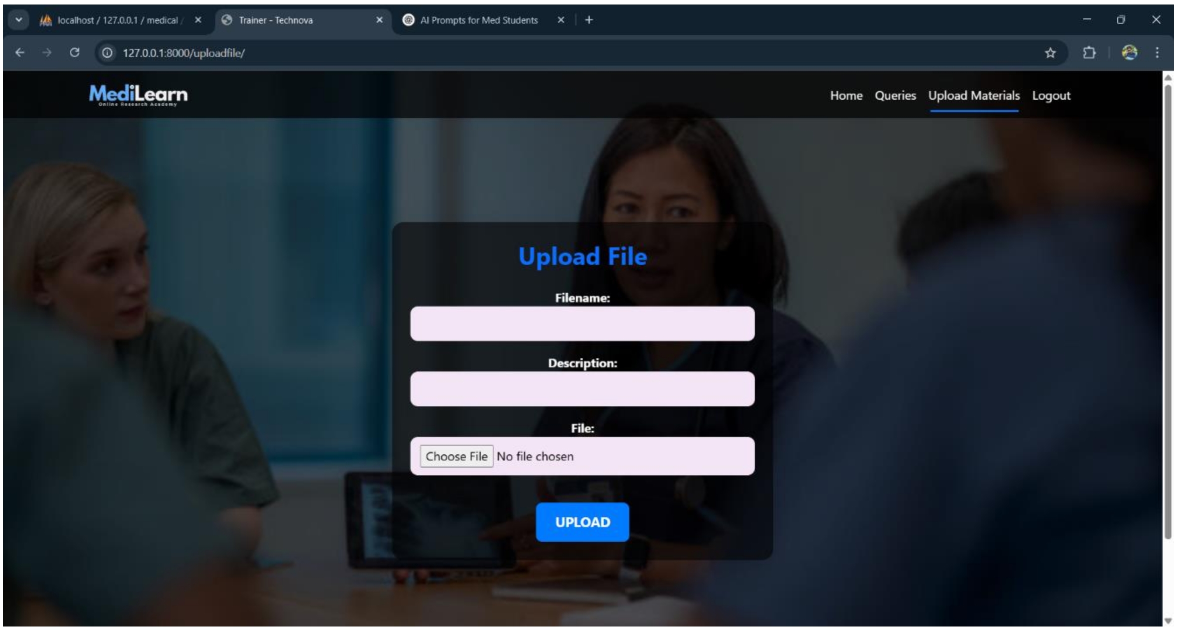
Home: Trainers can access their dashboard, which provides an overview of student interactions and uploaded materials. It serves as a control center for managing teaching activities efficiently.



Queries: Trainers can view and respond to student queries, providing clarifications and additional explanations. This feature ensures continuous support and engagement in the learning process.

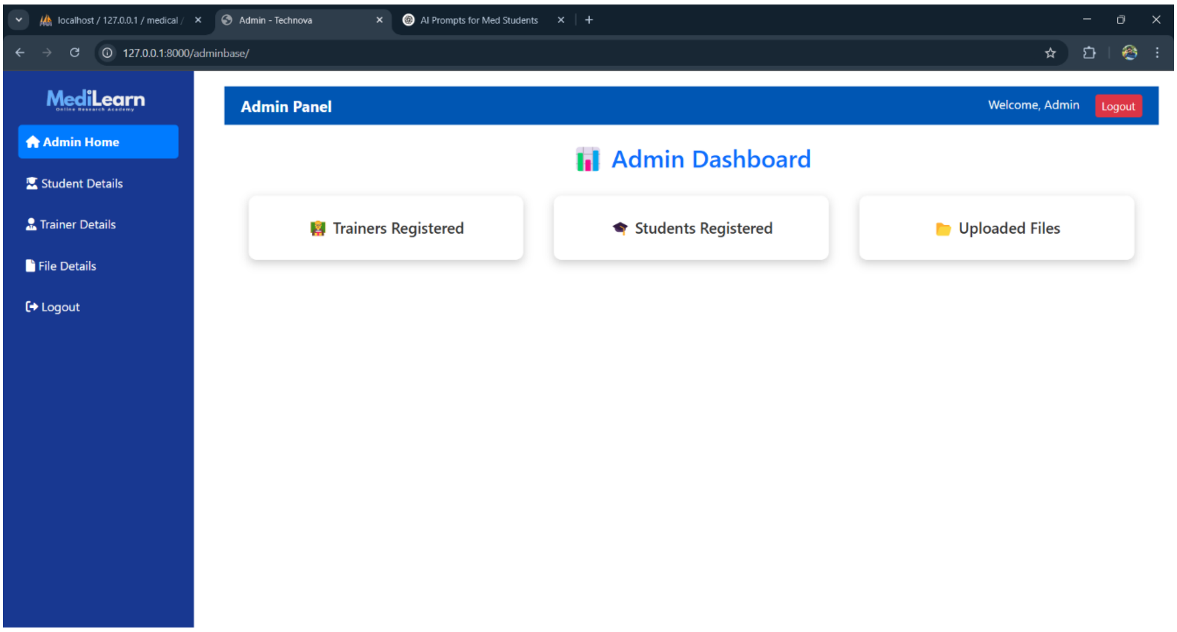


Upload Materials: Trainers can upload study resources, including PDFs, videos, and assignments. This allows them to update content based on evolving curriculum requirements.

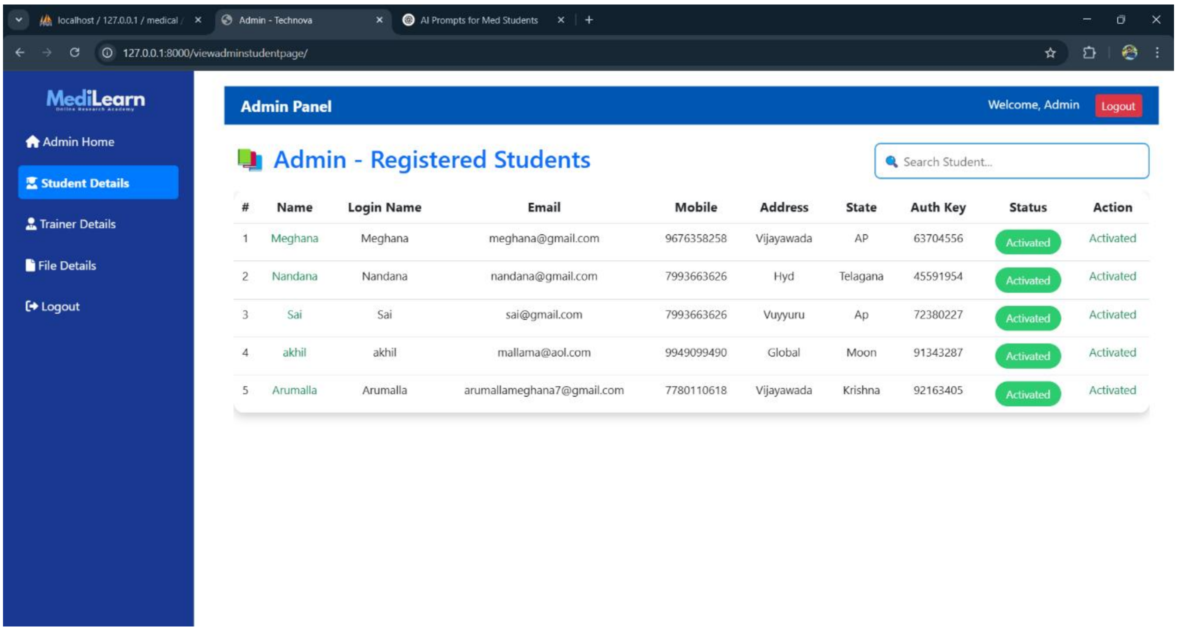


ADMIN MODULE

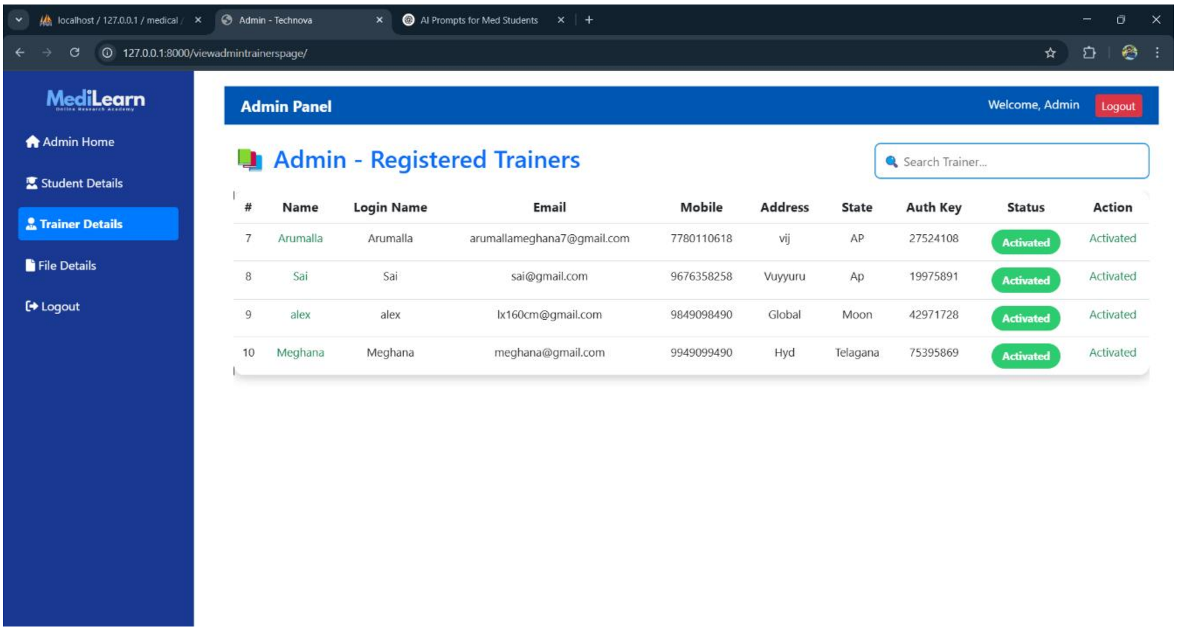
Admin Home: The admin dashboard provides an overview of platform activities, including student and trainer registrations. It allows administrators to monitor system performance and ensure smooth operations.



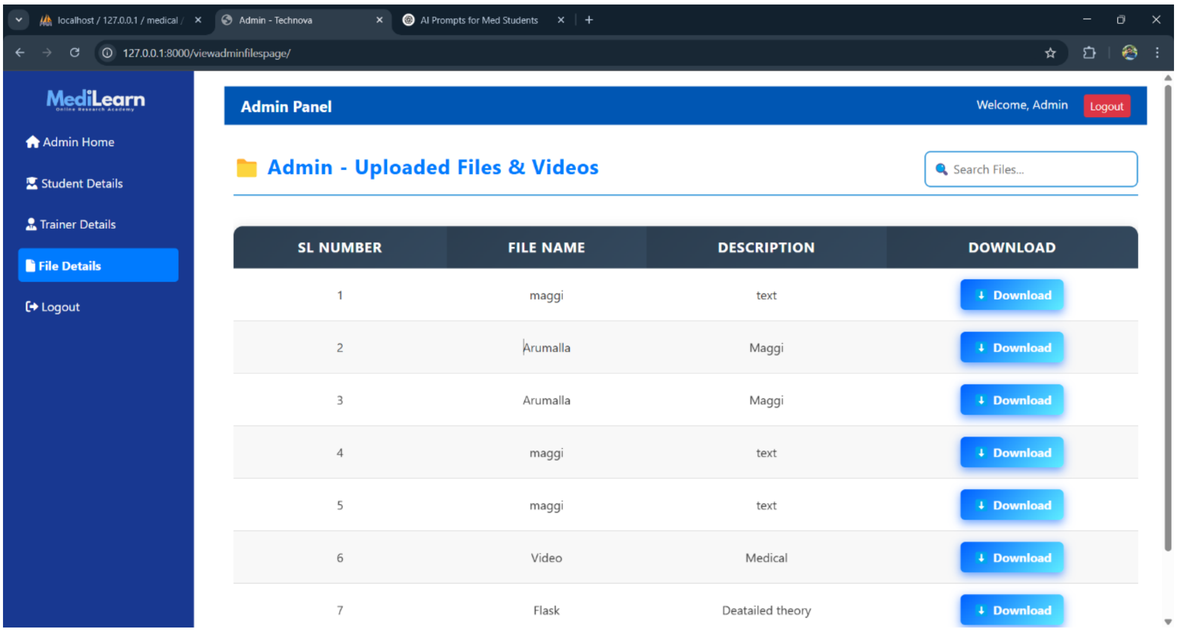
Student Registered Details: Admins can view and maintain student registration records. They ensure that all registered students have proper access to courses and system features.



Trainer Registered Details: Admins can access and manage trainer registration details. This includes verifying credentials and approving trainer accounts for platform access.



File Details: Admins can manage and track uploaded files, ensuring content organization and compliance. They can also remove outdated or irrelevant materials from the system.



8. CONCLUSION

The essence of education is accumulation and inheritance, inheriting the knowledge accumulated by the predecessors to future generations and encouraging them to innovate through educational means. The fundamental of artificial intelligence technology is to accumulate knowledge through machine learning, artificial neural network, data mining and other methods. Through decision supports, the expert system spreads knowledge and applies it. This article analyzes the changes in the way that artificial intelligence technology modifies traditional medical education. A key way that artificial intelligence affects medical education is to support personalized learning, help students at different levels, and provide help and support when teachers and students need it. Artificial intelligence can not only help teachers and students design courses that meet their needs, but also can focus on student performance and alert teachers when problems may arise, helping teachers improve teaching methods. Artificial intelligence will change the role of teachers. Teachers will supplement artificial intelligence courses to provide students with interpersonal interaction and practical experience. Using artificial intelligence systems, students can learn anytime, anywhere, and some classroom teaching can be substituted by these programs.

9. FUTURE WORK

Building upon the current findings, future research can focus on integrating artificial intelligence with immersive technologies such as Virtual Reality (VR) and Augmented Reality (AR). These technologies, combined with AI, can offer medical students highly interactive and realistic simulations for surgical procedures, anatomy exploration, and emergency care training. This would enhance practical learning experiences beyond traditional classroom settings.

Another promising direction is the development of personalized learning systems powered by AI. These systems can adapt to each student's learning style, pace, and progress by analyzing performance data and customizing the educational content accordingly. Such an approach can increase student engagement and improve learning outcomes by providing targeted support.

In addition, intelligent tutoring systems can be explored further. These AI-powered tutors can provide one-on-one guidance, answer student queries, assess comprehension in real time, and offer instant feedback. This would help bridge the gap between self-study and instructor-led teaching, making learning more effective and accessible.

Natural Language Processing (NLP) is another area with vast potential. Future work could focus on using NLP to automatically transcribe and summarize medical lectures, analyze complex medical literature, and even generate quiz questions or case studies. This could greatly reduce the workload for educators while enhancing resource accessibility for students.

Finally, AI can be employed to analyze student data and predict academic performance. Predictive models could help identify students at risk of underperforming, allowing educators to provide timely interventions and personalized support. Overall, continued research and development in these areas will contribute to a more intelligent, inclusive, and effective medical education system.

10. REFERENCES

[1] Joshi S. Use of Information Technology in Medical Education [J]. Webmed Central Medical Education,2010,1(9): WMC00607.

[2] Johnson L, Adams Becker S, Cummins M, et al. NMC Horizon Report 2016 Higher Education Edition [R]. Texas: The NewMedia Consortium,2016.

[3] Thomas H. Learning Spaces, Learning Environments and Displacement of Learning [J]. British Journal of Educational Technology,2010,41 (3): 502-511.

[4] Youmei Wang. Cultivating Makers: Educational Ecosystem in the View of New Industrial Revolution [J]. Open Education Research, 2015, (5): 40, 49-56.

[5] Xinmin Sang, Yangbin Xie. Improving the Quality of University Teaching in Learning Innovation: The Key to Apply Information Technology in Higher Education [J]. Journal of Higher Education, 2012 (5): 64-69.